

Beneficial AI and Machine Ethics

6.1 INTRODUCTION

How should we direct AIs to promote human values? As we continue to develop powerful AI systems, it is crucial to ensure they are safe and beneficial for humanity. In this chapter, we discuss the challenges of specifying appropriate values for AI systems to pursue. Some of these questions are already relevant for AI systems that exist today in the healthcare or automotive sectors, where artificial systems may be making decisions in situations that can harm or benefit humans. They will become even more critical for future systems that may be integrated more broadly across economies, governments, and militaries, making high-stakes decisions with many important moral considerations. As discussed in section 3.4, these questions are core components of the overall challenge of “AI alignment”.

Many people have incoherent views on embedding values into AIs. People often talk about what AIs should do to promote human values. They may agree with many of the following:

1. AIs should do what you tell them to do.
2. AIs should promote what you choose to do.
3. AIs should do what’s fair.
4. AIs should do what a democratic process tells them to do.
5. AIs should figure out what is moral, then do that.
6. AIs should do what is objectively good for you.
7. AIs should do what would make people happy.

All of these seem like reasonable answers. At least at first glance, these all seem like excellent goals for AIs. However, many of these are incompatible, because they make *different normative assumptions* and *require different technical implementations*. Other suggestions such as “AIs should follow human intentions” are highly

vague. Put straightforwardly, while these sound attractive and similar, they are not the same thing.

This should challenge the notion that this is a straightforward problem with a simple solution, and that the real challenges lie only elsewhere, such as in making AI systems more capable. Those who believe there are easy ways to ensure that AIs act ethically may find themselves grappling with inconsistencies and confusion when confronted with instances in which their preferred methods appear to yield harmful behavior.

In this chapter, we will explore these issues, attempting to understand which answers, if any, take us closer to creating safe and beneficial AIs. We start by considering some goals commonly proposed to ensure that AI systems are beneficial for society. Current and future AI systems will need to comply with existing laws and to avoid biases and unfairness towards certain groups in society. Beyond this, many want AI to make our economic systems more efficient and to boost economic growth. We briefly consider the attractions of these goals, as well as some shortcomings:

1. **Law:** Should AI systems just be made to follow the law? We examine whether we can design AIs that follow existing legal frameworks, considering that law is a legitimate aggregation of human values that is time-tested and comprehensive while being both specific and adaptable. We will lay out the challenges of the occasional silence, immorality, or unrepresentativeness of established law.
2. **Bias and Fairness:** How can we design AI systems to be fair? We explore bias and fairness in AI systems and the challenges associated with ensuring outcomes created by AIs are fair. We will discuss different definitions of fairness and see how they are incompatible, as well as consider approaches to mitigating biases.
3. **Economic Engine:** Should we let the economy decide what AIs will be like? We consider how economic forces are shaping AI development, and why we might be concerned about letting economic incentives alone determine how AI is developed and which objectives AI systems pursue.

Machine ethics. While all of these proposals seem to capture important intuitions about what we value, we argue that they face significant limitations and are not sufficient on their own to ensure beneficial outcomes. We consider what it would mean to create AI systems that aim directly to improve human wellbeing. This provides a case study in thinking about how we can create AI systems that can respond appropriately to moral considerations, an emerging field known as machine ethics. While this is a highly ambitious goal, we believe it is likely to become increasingly relevant in coming years. As AI capabilities improve, they may be operating with an increasing level of autonomy and a broadening scope of potential actions they could take. In this context, approaches based only on compliance with the law or profit maximization are likely to prove increasingly inadequate in providing sufficient guardrails on AI systems' behavior. To specify in greater detail how they should react in a particular situation, AI systems will need to be able to identify and respond appropriately to the relevant values at stake, such as human wellbeing.

Wellbeing. If we are to set wellbeing or other values as goals for AI systems, one fundamental question that faces us is how we should specify these values. In the second part of this chapter, we consider various interpretations of wellbeing and how attractive it would be to have AI systems pursue these, assuming that they became capable enough to do this. We start by introducing several competing theories of wellbeing, which might present different goals for ethical AI systems. We examine theories of wellbeing that focus on pleasure, objective goods, and preference satisfaction. We then explore preference satisfaction in more detail and consider what kind of preferences AI systems should satisfy. AI systems can be created to satisfy individual preferences, but which preferences they should focus on is an open question. We focus on the challenges of deciding between revealed, stated, and idealized preferences. Next, we turn to consider whether AI systems should aim to make people happy and how we might use AIs to promote human happiness. Finally, we consider the challenge of having AIs maximize wellbeing not just for an individual, but across the whole of society. We look at how to aggregate total wellbeing across society, focusing on social welfare functions. We discuss what social welfare functions are and how to trade-off between equity and efficiency in a principled way.

Moral uncertainty. There are many cases where we may feel uncertain about what the right response is to a particular situation due to conflicting moral considerations. We consider how AI systems might respond to such situations. In the case of AI systems, we may deliberately want to introduce uncertainty into their reasoning to avoid over-confident decisions that could be disastrous from some moral perspectives. One option to address moral uncertainty is using a moral parliament, where ethical decisions are made by simulating democratic processes.

6.2 LAW

Why not have AIs follow the law? We have just argued that for AI systems to be safe and beneficial, we need to ensure they can respond to moral considerations such as wellbeing. However, some might argue that simply getting AIs to follow the law is a better solution.

The law has three features that give it an advantage over ethics as a model for safe and beneficial AI. In democratic countries, the law is a *legitimate* representation of our moral codes: at least in theory, it is a democratically endorsed record of our shared values. Law is *time-tested* and *comprehensive*; it has developed over many generations to adjudicate the areas where humans have consequential disagreements. Finally, legal language can be *specific* and *adaptable* to new contexts, comparing favorably to ethical language, which can often be interpreted in diverging ways.

The next subsection will expand on these features of the law. We will see how these features contrast favorably with ethics before arguing that we do need ethics after all, but alongside the law.

6.2.1 The Case For Law

Legitimate Aggregation of Values

In a democratic country, the law is influenced by the opinions of the populace. Democratic citizens vote on new laws, directly or through representatives. If they don't like part of an existing law, they have a range of legal means, such as advocating, protesting, and voting, to change it. Even though the law at any given time won't perfectly reflect the values of the citizenry, the method of arriving at law is usually *legitimate*. In other words, the citizens have input into the law.

Legitimacy provides the law with a clear advantage over ethics. The law provides a collection of rules and standards that enable us to differentiate illegal from legal actions. Ethics, on the other hand, isn't standardized or codified. To determine ethical and unethical actions, we have to either pick an ethical theory to follow, or decide how to weigh the differing opinions of multiple theories that we think might be true. But any of these options are likely to be more controversial than simply following the law. Ethics has no in-built method of democratic agreement.

However, just following the law isn't a perfect solution: there will always be an act of interpretation between the written law and its application in a particular situation. There is often no agreement over the procedure for this interpretation. Therefore, even if AI systems were created in a way that bound them to follow the law, a legal system with human decision-makers might have to remain part of the process. The law is only legitimate when interpreted by someone democratically appointed or subject to democratic critique.

Time-Tested and Comprehensive

Systems of law have evolved over generations. With each generation, new people are given the job of creating, enforcing, and interpreting the law. The law covers a huge range of issues and incorporates a wide range of distinctions. Because these bodies of law have been evolving for so long, the information encoded in the law is a record of what has worked for many people and is often considered an approximation of their values. This makes the law a particularly promising resource for aligning AI systems with the values of the population.

Laws are typically written without AIs in mind. Most laws are written with humans in mind. As a result, they assume things such as intent—for example, American law holds it a crime to knowingly possess a biological agent that is intended for use as a weapon. Both 'knowingly' and 'intended' are terms we may not be able to apply to AIs, since AIs may not have the ability to know or intend things. Having AIs follow laws that were written without AIs in mind might result in unusual interpretations and applications of the law.

Another area of ambiguity is copyright law. AIs are trained on a vast corpus of training data, created by developers, and ultimately run by users. When AIs create content, it is unclear whether the user, developer, or creator of the training data

should be assigned the intellectual property rights. IP laws were written for humans and human-led corporations creating content, not AIs.

Rules and Standards

Naively, we might think of the law as a system of rules. “Law” seems almost synonymous with “rule” in our language. When we talk about “the laws of physics” or “natural laws” in general, we mean something rigid and inflexible—when X happens, Y will follow. This is partly true: inflexible rules are a part of the law. For instance, take the rule: “If someone drives faster than the speed limit of 70mph, they will be fined \$200.” In this case, there is an objective trigger (driving faster than 70mph) and a clear directive (a \$200 fine). There isn’t much room for the judge to interpret the rule differently. This gives the lawmaker predictable control over how the law will be carried out.

A law based on rules alone would be flawed. However, a fixed speeding rule would mean fining someone who was accelerating momentarily to avoid hitting a pedestrian and not fining someone who continued to drive at the maximum speed limit around a blind turn, creating a danger for other drivers. Rules are always over-inclusive (they will apply to some cases we would rather not be illegal) and under-inclusive (they won’t apply to all cases we would like to be illegal).

To remedy this problem, a law can instead be based on a standard. In the speeding case, a standard could be “when someone is driving unreasonably, they will be fined in proportion to the harm they pose.” A judge could apply this standard to get the correct result in both cases above (speeding to avoid an accident and going full speed around a blind turn). Standards have their own problems: with standards rather than rules, the judge is empowered to interpret the standards based on their own opinion, allowing them to act in ways that diverge from the lawmaker’s intentions.

The law uses rules and standards. Using rules and standards alongside each other, the law can find the best equilibrium between carrying out the lawmaker’s intentions and accounting for situations they didn’t foresee [294]. This gives the law an advantage in the problem of maintaining human control of AI systems by displaying the right level of ambiguity.

Law is less ambiguous than ethical language, which can be very ambiguous. Phrases like “do the right thing”, “act virtuously” or “make sure you are acting consistently” can mean different things to different people. In contrast, it is more flexible than programming languages, which are brittle and designed to only fit into particular contexts. Legal language can maintain a middle ground between rigid rules and more sensible standards.

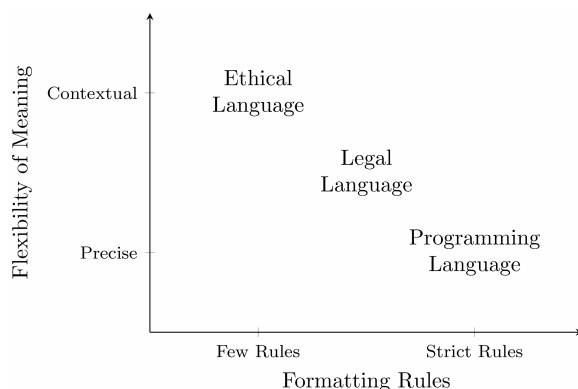


FIGURE 6.1. Legal language balances being less ambiguous than ethical language while being more flexibly formatted than programming language.

Specific and Adaptable

We can apply these insights about rules and standards in law to two core problems in controlling AI systems: *misinterpretation* and *gaming*. The law is specific enough to avoid misinterpretation and adaptable enough to prevent many forms of gaming.

Given commands in natural language, AIs might interpret them literally.

The misinterpretation problem arises when the literal interpretation of our commands differs from our intended interpretation. In AI safety discourse, we see this problem raised by many thinkers; for example, Stuart Russell raised the concern that an AI, if asked to develop a cure for cancer, might experiment on people, giving them tumors as part of carrying out the request [54]. A narrow, literal interpretation of “develop a cure for cancer,” which doesn’t take any of our typical considerations into account, could lead to this outcome. Misinterpretation risks are like wishes to a genie: we might get what we ask for but not what we want.

The capabilities of LLMs give us some reason to see misinterpretation risks as unlikely.

Before LLMs, some AI researchers were concerned that most AI models would be trained in ways that would mean they had no understanding of many ordinary human concepts or values. This might mean that we should be worried about extreme actions emerging from a misinterpretation of perfectly normal and mundane requests. However, this now seems less likely. Our experience with large language models has shown that by being trained on human-generated language data, AIs can respond to the meaning of our sentences in a way that is approximately similar to the way that a human speaker of the language would; for instance, the Happiness section below discusses two other systems that can predict human responses to video and text. If similar systems are used in the future, it seems plausible that AIs can apply laws in a sensible way.

That said, we might still worry that while AIs might usually interpret human concepts sensibly, they may still have representations that have rare quirks. This poses problems; for instance, it creates vulnerabilities that can be exploited by adversarial

attacks. There is more work to be done before we can feel comfortable that systems will reliably be able to interpret laws and other commands in practice.

However, we still face the risk of AIs gaming our commands to get what they want. Stuart Russell raises a different concern with AI: gaming [54]. An AI system may ‘play’ the system or rules, akin to strategizing in a game, to achieve its objectives in unexpected or undesired ways. He gives the example of tax codes. Humans have been designing tax laws for 6000 years, yet many still avoid taxation. Creating a tax code not susceptible to gaming is particularly difficult because individuals are incentivized to avoid paying taxes wherever possible. If our track record with creating rules to constrain each other is so bad, then we might be pessimistic about constraining AI systems that might have goals that we don’t understand.

A partial solution to misinterpretation and gaming is found in rules and standards. If we are concerned about misinterpretation, we might choose to rely on laws that are specific. Rules such as “any generated image must have a digital watermark” are specific enough that they are difficult to misinterpret. We might prefer using such laws rather than relying on abstract ethical principles, which are vaguer and easier to misinterpret.

Conversely, if we are concerned about AIs gaming rules, we might prefer to have standards, which can cover more ground than a rule. A well-formulated standard can lead to an agent consistently finding the right answer, even in new and unique cases. Such approaches are sometimes applied in the case of taxes. In the UK, for example, there is a rule against tax arrangements that are “abusive.” This is not an objective trigger: it is up to a judge to decide what is “abusive.” An AI system trained to follow the law can be accountable to rules and standards.

6.2.2 The Need for Ethics

This subsection will discuss why ethics is still indispensable in creating safe and beneficial AI, even though law is a powerful tool. Firstly, though the law is comprehensive, there are important areas of human life where it gives no advice. Secondly, it is common for laws to be immoral, even by the standards of the residents of the country that made them. Finally, the law is unlikely to be the most legitimate conceivable system, even if it is our best one.

Silent Law

The law is a set of constraints, not a complete guide to behavior. The law doesn’t cover every aspect of human life, and certainly not everything we need to constrain AI. Sometimes, this is accidental, but sometimes, the law is intentionally silent. We can call these zones of discretion: areas of behavior that the law doesn’t constrain; for example, a state’s economic policy and the content of most contracts between private individuals. The law puts some limits on these areas of behavior. Still, lawmakers intentionally leave free space to enable actors to make their own choices and to avoid the law becoming too burdensome.

The law often stops short of compelling us to do clearly good things. It is generally seen as good if someone steps in to rescue a stranger who is in danger. However, in some jurisdictions like the US, the duty to rescue only applies in certain cases where we have special responsibility for someone. We face a penalty if we fail to rescue our spouse or employee, but there is no law against failing to rescue a stranger, even when it would be at no cost to us. Giving humans this kind of discretion may not have terrible outcomes, because many people have strong ethical intuitions anyway, and would carry out the rescue. But an AI that was only following the law would not unless it had a strong ethical drive as well. In cases like rescue, where AI systems may be *more* capable than humans to help, we could be passing up a major benefit by asking the AI to only follow the law. Likewise, in the US, AIs may be required to avoid uttering copyrighted content and libelous claims, but doxing, instructions for building bombs, or advice for how to break the law can be legal, even if it is not ethical.

Conversely, by constraining AI with laws rather than guiding it with ethics, we risk it acting in undesirable ways in zones of discretion. AI could recommend potentially harmful economic policies, trick humans into regrettable contracts, and pursue legal but harmful business practices. Without ethics, a law-abiding AI could carry out great harm because the law is silent in situations where we may want to guide behavior nonetheless.

Immoral Law

The law can be immoral even if it is created legitimately and interpreted correctly. Democratic majorities can pass laws that a large number of fellow citizens think are immoral, or that will seem immoral to future populations of that country, such as the legalization of wars later regretted, legal slavery, legal discrimination on the basis of race and gender, and various other controversial laws which are later deemed morally wrong. There is also often a significant time delay between the moral opinions of a population changing, and the law changing to reflect them [295]. This time delay can be especially harmful in the context of AIs since they can create several new legal issues relatively quickly. This means that laws formed in democracies can fall short of being moral, even in the eyes of the citizens of the country that made them.

On many issues, the boundaries of the law are surprising. Since laws are often developed ad hoc, with new legislation to tackle the issues of the day, what laws actually permit and prohibit can often be unintuitive. For example, the ‘Anarchist’s Cookbook’, released in the US during the Vietnam War, had detailed instructions on how to create narcotics and explosives and sold millions of copies without being taken out of circulation because the law permitted this use of mass media. Similarly, “doxing” someone with publicly available information and generating compromising images and videos of real people are legal as well. Such acts seem intuitively illegal, but US law permits them.

If solely constrained by a country's fallible democratic laws, future AI systems could behave in ways most of the world would consider immoral. To mitigate this risk, it may be necessary to train AI systems to respect the ethical perspectives of those affected by their actions.

Unrepresentative Law

Law isn't the only way, or even necessarily the best way, to arrive at an aggregation of our values. Not all judges and legal professionals agree that the law *should* capture the values of the populace. Many think that legal professionals know better than the public, and that laws should be insulated from the changing moral opinions of the electorate. This suggests that we might be able to find, or conceive of, more representative ways of capturing the values of the populace. Current alternatives like surveys or citizens' assemblies are useful for some purposes, such as determining preferences on specific issues or arriving at informed, representative policy proposals. However, they aren't suited to the general task of summarizing the values of the entire population across areas as comprehensive as those covered by the law.

A Note On The Three Laws of Robotics

Some propose that Isaac Asimov's "Three Laws of Robotics" provide a useful set of rules for creating AIs that behave ethically [296, 297]. These laws are as follows:

- 1) "A robot may not injure a human being, or through inaction, allow a human being to come to harm,"
- 2) "A robot must obey the orders given to it by a human being except where such orders would conflict with the First Law," and
- 3) "A robot must protect its own existence as long as such protection does not conflict with the First or Second Laws."

These laws are not meant to provide a solution to ethical challenges. Asimov himself frequently tested the adequacy of these laws throughout his writing, showing that they are, in fact, limited in their ability to resolve ethical problems. Below, we explore some of these limitations.

Asimov's laws are insufficient for guiding ethical behavior in AI systems [298]. The three laws use under-defined terms like "harm" and "inaction." Because they're under-defined, they could be interpreted in multiple ways. It's not clear precisely what "harm" means to humans, and it would be even more difficult to encode the same meaning in AI systems.

Harm is a complex concept. It can be physical or psychological. Would a robot following Asimov's first laws be required to intervene when humans are about to hurt each other's feelings? Would it be required to intervene to prevent a human from behaving in ways that are self-harming but deliberate, like smoking? Consider the case of amputating a limb in order to stop the spread

of an infection. An AI programmed with Asimov's laws would be forbidden from amputating the limb, as that would literally be an instance of injuring a human being. However, the AI would also be forbidden from allowing the harmful spread of an infection through its inaction. These scenarios illustrate that the first law fails, and therefore, that the following two do not follow. The laws may need to be much more specific in order to reliably guide ethical behavior in future AI systems.

Philosophy has yet to produce a sufficient set of rules to determine moral conduct. The safety of future AI systems cannot be guaranteed simply through a set of rules or axioms. Numerous factors, such as proxy gaming and competitive pressures, cannot be adequately captured in a set of rules. Rules may be useful, but AI safety will require a more dynamic and comprehensive approach that can address existing technical and sociotechnical issues.

Overall, Asimov's Three Laws of Robotics fail to reliably guide ethical behavior in AI systems, even if they serve as a useful starting point for examining certain questions and problems in AI safety.

Conclusions About Law

The law is comprehensive, but not comprehensive enough to ensure that the actions of an AI system are safe and beneficial. AI systems must follow the law as a baseline, but we must also develop methods to ensure that they follow the demands of ethics as well. Relying solely on the law would leave many gaps that the AI could exploit, or make ethical errors within. To create beneficial AI that acts in the interests of humanity, we need to understand the ethical values that people hold over and above the law.

6.3 FAIRNESS

We can use the law to ensure that AIs make fair decisions. AIs are being used in many sensitive applications that affect human lives, from lending and employment to healthcare and criminal justice. As a result, unfair AI systems can cause serious harm.

The COMPAS case study. A famous example of algorithmic decision-making in criminal justice is the COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) software used by over 100 jurisdictions in the US justice system. This algorithm uses observed features such as criminal history to predict recidivism, or how likely defendants are to reoffend. A ProPublica report [299] showed that COMPAS disproportionately labeled African-Americans as higher risk than white counterparts with nearly identical offense histories. However, COMPAS's creators argued that it was *calibrated*, with accurate general probabilities of recidivism across the three general risk levels, and that it was less biased and better than human judg-

ments [300]. This demonstrated the trade-off between different definitions of fairness: it was calibrated across risk levels, but it was also clear that COMPAS generated more false positives for African-American defendants (predicting they would re-offend when they did not) and more false negatives for white defendants, predicting they would not re-offend when they in fact did. Adding to the concern, COMPAS is a black-box algorithm: its process is proprietary and hidden. One lawsuit argued this violates due process rights since its methods are hidden from the court and the defendants [301]. In this section, we will discuss some of the serious ethical questions raised by this case, examining what makes algorithms unfair and considering some methods to improve fairness.

6.3.1 Bias

AI systems can amplify undesirable biases. AI systems are being increasingly deployed throughout society. If these influential systems have biases, they can reinforce disparities and produce widespread, long-term harms. In AI, *bias* refers to a consistent, systematic, or undesirable distortion in the outcomes produced by an AI system. These outcomes can be predictions, classifications, or decisions. Bias can be influenced by many factors, including erroneous assumptions, training data, or human biases. Biases in modern deep learning systems can be especially consequential. While not all forms of bias are harmful, we focus on biases that are socially relevant because of their harms. We must proactively prevent bias to avoid its harms. This section overviews bias in AI and outlines some mitigation strategies.

Aspects of bias in AI. A bias is *systematic* when it includes a pattern of repeated deviation from the true values in one direction. Unlike random unstructured errors, or “noise,” these biases are not reliably fixed by just adding more data. Resolving ingrained biases often requires changing algorithms, data collection practices, or how the AI system is applied. *Algorithmic bias* occurs when any computer system consistently produces results that disadvantage certain groups over others. Some biases are relatively harmless, like a speech recognition system that is better at interpreting human language than whale noises. However, other forms of bias can result in serious social harms, such as partiality to certain groups, inequity, or unfair treatment.

Bias can manifest at every stage of the AI lifecycle. From data collection to real-world deployment, bias can be introduced through multiple mechanisms at any step in the process. Historical and social prejudices produce skewed training data, propagating flawed assumptions into models. Flawed models can cement biases into the AI systems that help make important societal decisions. In addition, humans misinterpreting results can further compound bias. After deployment, biased AI systems can perpetuate discriminatory patterns through harmful feedback loops that exacerbate bias. Developing unbiased AI systems requires proactively identifying and mitigating biases across the entire lifecycle.

Biases in AI often reflect systemic biases. Systematic biases can occur even against developers’ intentions. For instance, Amazon developed an ML-based resume-

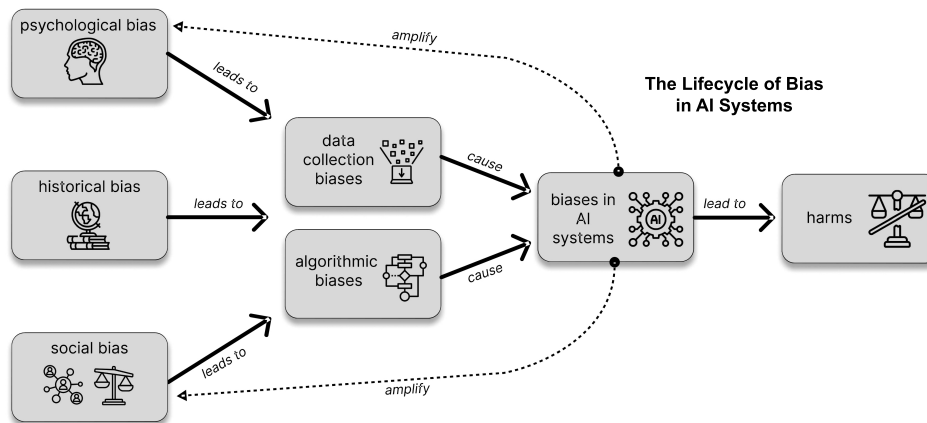


FIGURE 6.2. Systematic psychological, historical, and social biases can lead to algorithmic biases within AI systems.

screening algorithm trained on historical hiring decisions. However, as the tech industry is predominantly male, this data reflected skewed gender proportions in the data (about 60% male and 40% female) [302]. Consequently, the algorithm scored male applicants higher than equally qualified women, penalizing resumes with implicit signals like all-female colleges. The algorithm essentially reproduced real-world social biases in hiring and employment. This illustrates how biased data, when fed into AI systems, can inadvertently perpetuate discrimination. Organizations must be vigilant about biases entering any stage of the machine learning pipeline.

In many countries, some social categories are legally protected from discrimination. Groups called *protected classes* are legally protected from harmful forms of bias. These often include race, religion, sex/gender, sexual orientation, ancestry, disability, age, and others. Laws in many countries prohibit denying opportunities or resources to people solely based on these protected attributes. Thus, AI systems exhibiting discriminatory biases against protected classes can produce unlawful outcomes. Mitigating algorithmic bias is crucial for ensuring that AI complies with equal opportunity laws by avoiding discrimination.

6.3.2 Sources of Bias

Biases can arise from multiple sources, both from properties of the AI system itself and human interaction with the system. This section discusses common sources of harmful biases in AI systems, although there are many more. First, we will discuss technical sources of bias, primarily from flawed data or objectives. Then, we will review some biases that arise from interactions between humans and AI systems.

Technical Sources of Bias in AI Systems

An overview of technical sources of bias. In this section, we will review some sources of bias in technical aspects of AI systems. First, we will investigate some *data-*

driven sources of biases, including flawed training data, subtle patterns that can be used to discriminate, biases in how the data is generated or reported, and underlying societal biases. Flawed or skewed training data can propagate biases into the model's weights and predictions. Then, we show how RL training environments and objectives can also reinforce bias.

ML models trained on biased datasets can learn and reinforce harmful societal biases. AI systems learn from human-generated data, absorbing both valuable knowledge and harmful biases. Even when unintentional, this data frequently mirrors ingrained societal prejudices. As a result, AI models can propagate real-world discrimination by learning biases from their input data. For instance, a lawsuit found that Facebook's ad targeting algorithm violated the Fair Housing Act because it learned to exclude users from seeing housing ads based on race, gender, or other protected traits. Similarly, ML models can reflect political biases, deprioritizing users from specific political affiliations by showing their content to smaller audiences. As another example, an NLP model trained on a large corpus of internet text learned to reinforce gender stereotypes, completing sentence structures of the format "man is to X as woman is to Y" with content such as "man is to computer programmer as woman is to homemaker" [303]. These examples show how ML models can amplify existing social biases.

Models can learn to discriminate based on subtle correlations. One intuitive way to fix bias is to remove protected attributes like gender and achieve "fairness through unawareness." But this is not enough to remove bias. ML models can learn subtle correlations that serve as proxies for these attributes. For example, even in datasets with gender information removed, resume-screening models learned to associate women with certain colleges and assigned them lower scores [302]. In another study, ML models erroneously labeled images of people cooking as women, due to learned gender biases [304]. Thus, models can discriminate even when the data does not contain direct data about protected classes. This hidden discrimination can harm protected groups despite efforts to prevent bias.

Biased or unrepresentative data collection can lead to biased decisions. Training data reflects biases in how the data was collected. If the training data is more representative of some groups than others, the predictions from the model may also be systematically worse for the underrepresented groups. Thus, the model will make worse or biased decisions for the group that is represented less in the dataset. Imbalances in training data occur when the data is skewed with respect to output labels, input features, and data structure. For instance, a disease prediction dataset with 100,000 healthy patients but only 10 sick patients exhibits a large class imbalance. The minority class with fewer examples is underrepresented.

Several other problems can introduce bias in AI training data. Systematic problems in the data can add bias. For instance, *reporting bias* occurs when the relative frequency of examples in the training data misrepresents real-world frequencies. Often, the frequency of outcomes in legible data does not reflect their actual

occurrence. For instance, the news amplifies shocking events and under-reports normal occurrences or systematic, ongoing problems—reporting shark attacks rather than cancer deaths. *Sampling bias* occurs when the data collection systematically over-samples some groups and undersamples others. For instance, facial recognition datasets in Western countries often include many more lighter-skinned individuals. *Labeling bias* is introduced later in the training process, when systematic errors in the data labeling process distort the training signal for the model. Humans may introduce their own subjective biases when labeling data.

Beyond problems with the training data, the training environments and objectives of RL models can also exhibit problems that promote bias. Now, we will review some of these sources of bias.

Training environments can also amplify bias. *Reward bias* occurs when the environments used to train RL models introduce biases through improper rewards. RL models learn based on the rewards received during training. If these rewards fail to penalize unethical or dangerous behavior, RL agents can learn to pursue immoral outcomes. For example, models trained in video games may learn to accomplish goals by harming innocents if these actions are not sufficiently penalized in training. Some training environments may fail to encourage good behaviors enough, while others can even incentivize bad behavior by rewarding RL agents for taking harmful actions. Humans must carefully design training environments and incentives that encourage ethical learning and behavior [305].

RL models can optimize for training objectives that amplify bias or harm. Reinforcement learning agents will try to optimize the goals they are given in training, even if these objectives are harmful or biased, or reflect problematic assumptions about value. For example, a social media news feed algorithm trained to maximize user engagement may prioritize sensational, controversial, or inflammatory content to increase ad clicks or watch time. Technical RL objectives often make implicit value assumptions that cause harm, especially when heavily optimized by a powerful AI system [58, 306]. News feed algorithms implicitly assume that how much a user engages with some content is a high-quality indicator of the *value* of that content, therefore showing it to even more users. After all, social media companies train ML models to maximize ad revenue by increasing product usage, rather than fulfilling goals that are harder to monetize or quantify, such as improving user experience or promoting accurate and helpful information. Especially when taken to their extreme and applied at a large scale, RL models with flawed training objectives can exacerbate polarization, echo chambers, and other harmful outcomes. Problems with the use of flawed training objectives are further discussed in section 3.3.

Biases from human-AI interactions

Interactions between humans and AI systems can produce many kinds of bias. It is not enough to just ensure that AI systems have unbiased training data: humans interacting with AI systems can also introduce biases during development, usage, and monitoring. Flawed evaluations allow biases to go unnoticed before models

are deployed. *Confirmation bias* in the context of AI is when people focus on algorithm outputs that reinforce their pre-existing views, dismissing opposing evidence. Humans may emphasize certain model results over others, skewing the outputs even if the underlying AI system is reliable. This distorts our interpretation of model decisions. *Overgeneralization* occurs when humans draw broad conclusions about entire groups based on limited algorithmic outputs that reflect only a subset. Irrationality and human cognitive bias play a substantial role in biasing AI systems.

Human-AI system biases can be reinforced by feedback loops. *Feedback loops* in human-AI systems often arise when the output of an AI system is used as input in future AI models. An AI system trained on biased data could make biased decisions that are fed into future models, reinforcing bias in a self-perpetuating cycle. We speak more about these feedback loops in chapter 5. *Self-fulfilling prophecies* can occur when an algorithmic decision influences actual outcomes, as the model reinforces its own biases and influences future input data [307]. In this way, models can amplify real-world biases, making them even more real. For example, a biased loan-approval algorithm could deny loans to lower-income groups, reinforcing real-world income disparities that are then reflected in the training data for future models. This process can make bias more severe over time.

Automation and measurability induce bias. Bias can be amplified by *automation bias*, where humans favor algorithmic decisions over human decisions, even if the algorithm is wrong or biased. This blind trust can cause harm when the model is flawed. Similarly, a *bias toward the measurable* can promote a general preference for easily quantifiable attributes. Human-AI systems may overlook important qualitative aspects and less tangible factors.

Despite their problems, AI systems can be less biased than humans. Although there are legitimate concerns, AI systems used for hiring and other sensitive tasks may sometimes lead to *less* biased decisions when compared with human decision-makers. Humans often harbor strong biases that skew their judgment in these decisions. With careful oversight and governance, AI holds promise to reduce certain biases relative to human motivations.

6.3.3 AI Fairness Concepts

Methods for improving AI fairness could mitigate harms from biased systems, but they require overcoming challenges in formalizing and implementing fairness. This section explores *algorithmic fairness*, including its technical definitions, limitations, and real-world strategies for building fairer systems.

Fairness is difficult to specify. Fairness is a complicated and disputed concept with no single agreed-upon definition. Different notions of fairness can come into conflict, making it challenging to ensure that an AI system will be considered fair by all stakeholders.

Five fairness concepts. Some concepts of *individual fairness* focus on treating similar individuals similarly—for instance, ensuring job applicants with the same qualifications have similar chances of being shortlisted. Others focus on *group fairness*: ensuring that protected groups receive similar outcomes as majority groups. *Procedural fairness* emphasizes improving the processes that lead to outcomes, making sure they are consistent and transparent. *Distributive fairness* concerns the equal distribution of resources. *Counterfactual fairness* emphasizes that a model is fair if its predictions are the same even if a protected characteristic like race were different, all else being equal. These concepts can all be useful in different contexts.

Justice as fairness. Ethics is useful for analyzing the idea of fairness. John Rawls' theory of justice as fairness argues that fairness is fundamental to achieving a more just social system. His *maximin* and *difference principles* state that inequalities in social goods can only be justified if they maximize benefits for the most disadvantaged people. He also argued that the social goods must be open to all under equality of opportunity. These ideas align with common notions of fairness. Some argue this principle also applies to AI: harms from the bias of algorithmic decisions should be minimized, especially in ways that make the worst-off people better off. Theories of justice can help develop the background principles for fairness.

Algorithmic fairness. The field of algorithmic fairness aims to understand and address unfairness issues that can arise in algorithmic systems, such as classifiers and predictive models. This field's goal is to ensure that algorithms do not perpetuate disadvantages based on *protected characteristics* such as race, gender, or class, especially while predicting an outcome from features based on training data. Several different technical definitions of fairness have been proposed, often formalized mathematically. These definitions aim to highlight unfairness in ML systems, but most possess inherent limitations. We will review three definitions below.

Statistical parity. The concept of statistical parity, also known as demographic parity, requires that an algorithm makes positive decisions at an equal rate for different groups. This metric requires that the model's predictions are *independent* of the sensitive attribute. A hiring algorithm satisfies statistical parity if the hiring rates for men and women are identical. While intuitive, statistical parity is a very simplistic notion; for instance, it does not account for potential differences between groups that could justify or explain different outcomes.

Equalized odds. Equalized odds require that the false positive rate and false negative rate are equal across different groups. A predictive health screening algorithm fulfills equalized odds if the false positive rate is identical for men and women. This metric ensures that the *accuracy* of the model is not dependent on the sensitive attribute value. However, enforcing equalized odds can reduce overall accuracy.

Calibration. Calibration measures how well predicted probabilities match empirical results. In a calibrated model, the actual long-run frequency of positives in the real population will match the predicted probability from the model. For instance, if the model predicts 20% of a certain group will default on a loan, roughly 20% will in fact default. Importantly, calibration is a metric for populations, and it does not tell us

about the correctness or fairness of an ML system for individuals. Calibration can improve fairness by preventing incorrect, discriminatory predictions. As it happens, ML models often train on losses that encourage calibration, and are therefore often calibrated naturally.

These technical concepts can be useful for operationalizing fairness. However, there is no single mathematical definition of fairness that matches everyone's complex social expectations. This is a problem because satisfying one definition can often violate others: there are tensions between statistical notions of fairness.

6.3.4 Limitations of Fairness

There are several problems with trying to create fair AI systems. While we can try to improve models' adherence to the many metrics of fairness, the three classic definitions of fairness are mathematically contradictory for most applications. Additionally, improving fairness is often at odds with accuracy. Another practical problem is that creating fair systems means different things across different areas of applications, such as healthcare and justice, and different stakeholders within each area have different views on what constitutes fairness.

Contradictions between fairness metrics. Early AI fairness research largely focused on three metrics of fairness: statistical/demographic parity, equalized odds, and calibration. However, these ubiquitous metrics often contradict each other: statistical parity only considers overall prediction rates, not accuracy, while an equalized odds approach focuses on accuracy across groups and calibration emphasizes correct probability estimates on average. Achieving calibration may require violating statistical parity when the characteristic being predicted is different across groups, such as re-offending upon release from prison being more common among disadvantaged minorities [308]. This makes fulfilling all three notions of fairness at once difficult or impossible.

The *impossibility theorem* for AI fairness proves that no classifier can satisfy these three definitions of fairness unless the prevalence of the target characteristic is equal across groups or prediction is perfect [309, 310]. Requiring a model to be “fair” according to one metric may actually disadvantage certain groups according to another metric. This undermines attempts to create a universally applicable, precise definition of fairness. However, we can still use metrics to better approximate our ideals of fairness while remaining aware of their limitations.

Fairness can reduce performance if not achieved carefully. Enforcing fairness constraints often reduces model accuracy. Two papers found that applying fairness techniques to an e-commerce recommendation system increased financial costs [311] and mitigating unfairness in Kaggle models by post-processing reduced performance [312]. However, these and others also find ways to simultaneously improve both fairness and accuracy; for example, work on healthcare models has managed to improve fairness with little effect on accuracy [313]. While aiming for fairness can

reduce model accuracy in many cases, sometimes fairness can be improved without harming accuracy.

Difficulties in achieving fairness across contexts. Different fields have distinct problems: fairness criteria that make sense in the context of employment may be inapplicable in healthcare. Even different fields within healthcare face different problems with incompatible solutions. These context-specific issues make generic solutions inadequate. Models trained on historical data might reflect historical patterns such as the underprescription of pain medication to women [314]. Removing gender information from the dataset seems like an obvious way to avoid this problem. However, this does not always work and can even be counterproductive. For instance, removing gender data from an algorithm that matches donated organs to people in need of transplants failed to eliminate unfairness, because implicit markers of gender like body size and creatinine levels still put women at a disadvantage [315]. Diagnostic systems without information about patients' sex tend to mispredict disease in females because they are trained mostly on data from males [316]. Finding ways to achieve fairness is difficult: there is no single method or definition of fairness that straightforwardly translates into fair outcomes for all.

Disagreements in intuitions about fairness. There is widespread disagreement in intuitions about the fairness of ML systems, even when a model fulfills technical fairness metrics; for instance, patients and doctors often disagree on what constitutes fairness. People often view identical decisions as more unfair if they come from a statistical model [317]; they also often disagree on which fairness-oriented features are the most important [318], such as whether race should be used by the model or whether the model's accuracy or false positive rates are more important. It is unclear how to define fairness in a generally acceptable way.

6.3.5 Approaches to Combating Bias and Improving Fairness

Due to the impossibility theorem and inconsistent and competing ideas, it is only possible to pursue some *definition* or *metric* of fairness—fairness as conceptualized in a particular way. This goal can be pursued both through technical approaches that focus directly on algorithmic systems, and other approaches that focus on related social factors.

Technical approaches. Metrics of fairness such as statistical parity identify aspects of ML systems that are relevant for fairness. Technical approaches to improving fairness include a host of methods to improve models' performance on these metrics, which can mitigate some forms of unfairness. These often benefit from being broadly applicable with little domain-specific knowledge. Developers can test predictive models against various metrics for fairness and adjust models so that they perform better. Fairness toolkits offer programmatic methods for implementing technical fairness metrics into ML pipelines. Other methods for uncovering hidden sources of unfairness in ML models include adversarial testing, sensitivity analysis, and ranking feature importances. One promising technical approach involves training an adversarial network

to predict a protected variable from an ML model's outputs [319]. By penalizing the model when the adversary succeeds at predicting a variable like race or political affiliation from the model's outputs, the model is forced to avoid discrimination and make predictions that do not unfairly depend on sensitive attributes. When applied well, this can minimize biases.

Problems with technical approaches. However, technical methods fall short of addressing the social consequences of unfairness. They fail to adjust to sociocultural contexts and struggle to combat biases inherited from training data. “Fairness through unawareness” aims to remove protected characteristics like gender and race from datasets to prevent sexism and racism, but often fails in practice because this data is embedded in correlates. A focus on narrow measures can ignore other relevant considerations, and measures are often subject to proxy gaming 3.3. A more in-depth, qualitative, and socially grounded approach is often harder and does not scale as easily as technical methods, but it is still essential for navigating concerns in AI fairness.

Engineering strategies for reducing bias must be paired with non-technical strategies. Ultimately, technical debiasing alone is insufficient. Social processes are crucial as humans adapt to AI. We speak about this at length in section 4.5, but here we will only mention a few ideas. For instance, *early bias detection* involves creating checks to identify risks of bias before the AI system is deployed or even trained, so that models that have discriminatory outputs can be rejected before they cause harm. Similarly, *gradual deployment* safely transitions AI systems into use while monitoring them for bias so that harms can be identified early and reversed. *Regulatory changes* can require effective mitigation strategies by law, mandating transparency and risk mitigation in safety-critical AI systems, as we discuss in chapter 8.

Other approaches. Other approaches emphasize that unfairness is tied to systemic social injustices propagated through technical systems. They highlight political, economic, and cultural factors and apply methods such as anti-discrimination policy, legal reform, and a design process focused on values and human impacts. These methods, which include policies like developing AI systems with input from stakeholders, can surface and mitigate sources of unfairness early. Substantive social changes are generally more expensive and difficult than technical approaches. However, they can be more impactful, reducing models' negative social impacts.

Participatory design can mitigate bias in AI development. An important non-technical strategy for bias reduction is stakeholder engagement, or deeply engaging impacted groups in the design of the AI system to identify potential biases proactively. Diverse teams and users can also help engineering teams incorporate diverse perspectives into the R&D process of AI models to anticipate potential biases proactively. One approach to proactively addressing bias is *participatory design*, which aims to include those affected by a developing technology as partners in the design process to ensure that the final product meets diverse human interests. For

example, before implementing an AI notetaking assistant for doctors, participatory design can require hospitals to improve the system based on feedback from all stakeholders during iterative design sessions with nurses, doctors, and patients. Rather than just evaluating ML models on test sets, developers should consult with the affected groups during the design process. Adding oversight mechanisms for rejecting models with discriminatory outputs can also enable catching biases before AI models affect real decisions.

Independent audits are important for identifying biases in AI systems before deployment. Auditors can systematically evaluate datasets, models, and outputs to uncover discrimination and hold the developers of AI systems accountable. There are several signs of bias to look for when auditing datasets. For example, auditors can flag missing data for certain subgroups, which indicates underrepresentation. *Data skew*, where certain groups are misrepresented compared to their real-world prevalence, is another sign of bias. Patterns and correlations with protected classes could indicate illegal biases. Auditors can also check for disparities in the model outputs. By auditing throughout the process, developers can catch biases early, improving data and models *before* their biases propagate. Rather than waiting until after the system has harmful impacts, meticulous audits should be integrated as part of the engineering and design process for AI systems [320]. Audits are especially effective when conducted independently by organizations without a stake in the AI system's development, allowing for impartial and rigorous auditing throughout the process.

Effective model evaluation is a crucial way to reduce bias. An important part of mitigating bias is proactively evaluating AI systems by analyzing their outputs for biases. Models can be tested by measuring performance metrics such as false positive and false negative rates separately for each subgroup. For instance, significant performance disparities between groups like men and women can reveal unfair biases. Ongoing monitoring across demographics is necessary to detect unintended discrimination before AI systems negatively impact people's lives. Without rigorous evaluation of model outputs, harmful biases can easily go unnoticed.

Reducing toxicity in data aims to mitigate harmful biases in AI, but faces challenges. *Toxicity* refers to harmful content, such as inflammatory comments or hate speech. Models trained on unfiltered text can absorb these harmful elements. As a result, AI models can propagate toxic content and biases if not carefully designed. For example, language models can capture stereotypical and harmful associations between social groups and negative attributes based on the frequency of words occurring together. Reducing toxicity in the training data can mitigate some biases. For example, developers can use toxicity classifiers to clean up the internet data, using both sentiment and manual labels to identify toxic content. However, all of these approaches still run into major challenges and limitations. Classifiers are still subject to social bias, evaluations can be brittle and unreliable, and bias is often very hard to measure.

Trade-offs can emerge between correcting one form of bias and introducing new biases. Bias reduction methods can introduce new biases, as classifiers have social biases, evaluations are unreliable, and bias reduction can introduce new biases. For example, some experiments show that an attempt to correct for toxicity in OpenAI’s older content moderation system resulted in biased treatment towards certain political and demographic groups: a previous system classified negative comments about conservatives as not hateful, while flagging the exact same comments about liberals as hateful [321]. It also exhibited disparities in classifying negative comments towards different nationalities, religions, identities, and more.

Conclusion

We have discussed some of the sources of bias in AI systems, including problems with training data, data collection processes, training environments, and flawed objectives that AI systems optimize. Human interactions with AI systems, such as automation bias and confirmation bias, can introduce additional biases.

We can clarify which types of bias or unfairness we wish to avoid using mathematical definitions such as statistical parity, equalized odds, and calibration. However, there are inherent tensions and trade-offs between different notions of fairness. There is also disagreement between stakeholders’ intuitions about what constitutes fairness.

Technical approaches to debiasing including predictive models and adversarial testing are useful tools to identify and remove biases. However, improving the fairness of AI systems requires broader sociotechnical solutions such as participatory design, independent audits, stakeholder engagement, and gradual deployment and monitoring of AI systems.

6.4 THE ECONOMIC ENGINE

What if we allow the economy to decide what AIs will be like? Unlike some prior technological breakthroughs (such as the development of nuclear energy and nuclear weapons), most investment in AI today is coming from businesses. Leading commercial AI developers have acquired the enormous computational resources required to train state-of-the-art systems and have hired many of the world’s best researchers. We could therefore argue that AI development is most closely aligned with business or economic goals such as wealth maximization. Many believe this is good, arguing that the development of AIs should be guided by market forces. If AI can accelerate economic growth, provided we can also ensure a fair distribution of costs and benefits of AI across society, this could be positive for people’s welfare around the globe. We examine the specific impacts of AI on economic growth and its distributional effects in the Governance chapter.

Here, we consider the broader attractions and limitations of allowing economic growth to be the main force determining how AI systems are developed. As part of this discussion, we will briefly introduce and explain a few basic concepts from economics,

such as market externalities, which provide an essential foundation for our analysis. We argue that while economic incentives can be powerful forces for prosperity and innovation, they do not adequately capture many important values. AI systems that are primarily created in order to maximise growth and profit for their developers could have a range of harmful side-effects. Alternative goals are discussed further in the following sections.

6.4.1 Allocative Efficiency of Free Markets

Competitive economic markets can drive efficiency and foster collective prosperity. Historically, market forces have led to specialization at a vast scale and worldwide competition, permitting the production of better goods and services at lower prices. The global economy is a complex system that allows for mass coordination. Prices, for instance, help producers and consumers find efficient trading equilibria in the face of dynamic market conditions. One can speculate that the integration of AI into the economic engine could further optimize production and enhance competition, ultimately contributing to accelerated economic growth and improved wellbeing.

Under the right conditions, the free market can also create allocative efficiency. The First Fundamental Theorem of Welfare Economics states that—subject to some strong assumptions—an equilibrium allocation of goods reached by trading on a free market must be *Pareto efficient*. An outcome is Pareto efficient if there is no way to make anyone better off without making someone else worse off: any change must be neutral or trade off one person’s welfare against another. Suppose the allocation of goods were not Pareto efficient, and two individuals could both be made better off. Then, the individuals would trade in a way that exploited the possible Pareto improvement to their mutual benefit.

This supports Adam Smith’s famous “Invisible Hand” argument, which suggests that when individuals pursue their self-interest within a market, they unintentionally contribute to social welfare. By creating gains from trade, there is an overall improvement in living standards for everyone.

There are many conditions required for the First Fundamental Theorem of Welfare Economics to hold:

1. **There must be an open market.** There should be no barriers to entry for producers or buyers, so that everyone can participate in this market. This openness stimulates competition, promoting economic efficiency.
2. **No seller should be big enough to move prices up alone.** If there are many sellers, then anyone who raises prices will lose consumers. There must be no monopoly power: anyone with the ability to raise prices without being forced down by competition will create distortions that leave consumers worse off.
3. **No producer should privately hold a pivotal technology.** This means that other producers should be able to copy the production of the first mover. While the first mover will make profits in the short run, the market allocation will be Pareto optimal in the long run.

4. **No buyer should be big enough to move prices down alone.** Similar to the condition for producers, no buyer should be big enough to force producers to take lower prices than others would offer for them. Such buyers would create distortions that might, for instance, force producers out of business, ultimately harming everyone.
5. **There must be perfect information for everyone.** Producers and consumers must have access to perfect information, such as about product quality and pricing. If consumers don't know, for instance, that a seller's product is defective or that other sellers are offering lower prices, then markets cannot achieve efficiency.
6. **There must be no externalities in consumption.** When one consumes a good, there must be no effect—positive or negative—on anyone else. Second-hand smoke has a negative externality on anyone nearby but, in a free market, this is unaccounted for in the price of cigarettes. As a result, the price does not reflect the total *social cost* of a cigarette: from a social wellbeing perspective, these are priced too low, creating inefficiency.
7. **Preferences must be non-satiated.** The first welfare theorem does not hold for all types of preferences; one technical restriction is that consumers should always prefer more of at least one good. If someone gains no further value from extra consumption, their trading equilibria may not be Pareto efficient.
8. **The state must enforce property and contract laws.** Most economic theory assumes that contracts are enforceable, and that individuals and corporations have protected property rights. Without these, trading would be difficult to achieve and much more costly, making it more difficult for everyone to achieve optimal outcomes.

Given the stringency of these conditions, it is obvious that they will not always hold in practice and that there may be market failures. Unregulated markets do not always create efficient outcomes: instead, unregulated markets often see informational asymmetries, market concentration, and externalities. Unfortunately, AIs may exacerbate these market failures and increase income and wealth inequality, creating disproportionate gains for the wealthy individuals and firms that own these systems while decreasing job opportunities for others.

6.4.2 Market Failures

In this section, we consider three common types of market failures that are especially relevant in the context of AI: *information asymmetries*, *oligopolies*, *externalities*. We also discuss the idea of *moral hazard*, which can be applied to the development of AI systems which can generate profit for their owners but could prove harmful to others.

Informational Asymmetries

Information known to only some can create market failures. Information asymmetry captures the idea that buyers and sellers have different information regarding the product they are trading. For instance, buyers are aware of the prod-

uct's quality and specifications, and sellers know their true willingness to pay for the product. Information asymmetry isn't inherently problematic—it is, in fact, often a positive aspect of market dynamics. We trust specialists to provide valuable services in their respective fields; for instance, we rely on our mechanics to know more about our car's inner workings than we do.

However, issues arise when information asymmetry leads to *adverse selection*. This occurs when sellers withhold information about product quality, leading buyers to suspect that only low-quality products are available. For example, in the used car market, a dealer might hide the fact that a car's axle is rapidly wearing out. Potential buyers, aware of this possibility but unable to distinguish good cars from bad, may only be willing to pay a low price that reflects the risk of buying a defective car. As a result, high-quality cars are driven out of the market, leaving only low-quality cars for sale. This leads to a market failure where all the high-quality goods are never traded, resulting in inefficiency [322]. Additionally, buyers who lack this crucial information might think that some cars are high-quality and pay accordingly, buying defective cars for high prices.

AIs may exacerbate informational asymmetries. AI holds the power to both create and exploit information asymmetries in unprecedented ways. This capacity can be employed in beneficial ways, like providing highly personalized services or excellent recommendations. However, it can also be misused, leading to situations in which those using AIs can manipulate consumers. AI-powered analytics allows companies to create sophisticated profiles of consumers that can uncover deep insights into individual personalities and behaviors. [323] Big tech companies already use social media and device activity to understand an individual's preferences and vulnerabilities better than ever before. This knowledge can be used to shape targeted advertisements or manipulations that are more effective. While these are not market failures in the technical sense, they can lead to exploitation and other undesirable outcomes.

Though AI may magnify the potential for information asymmetries, such strategies have long standing precedents in non-AI contexts. Predatory lending is a common practice where lenders, often equipped with more information than borrowers, use deceptive practices to encourage individuals into accepting unfair loan terms. These tactics tend to target lower-income and less-educated individuals who might not have the time or background to understand the fine-print of what they're signing, or the resources to find legal counsel. AIs can further increase the power imbalance; for instance, AIs might be used to predict who is most likely to accept these unfair loan terms based on their digital behavior, leading to even more targeted predatory lending. AIs can both amplify existing issues and present new challenges.

Oligopolies

Markets can tend towards concentration, leaving consumers worse off. While competition can create productive efficiency, some firms can avoid competitive pressures; for instance, utilities that are the exclusive provider of power or gas in a re-

gion might be able to raise their prices without losing many consumers to competitors due to lack of alternatives. When a single company or a small group of companies has a high level of control over a market, consumers are often left with limited product options and high prices. Markets in which a small number of companies are the only available suppliers are sometimes described as oligopolies. To preserve the benefits of competition, governments implement regulations such as antitrust laws, which they can use to limit market concentration by preventing mergers between companies that would give them excessive market power.

High initial investment requirements can impede newcomers from entering the market. Historically, first-movers in capital-intensive industries have a competitive advantage, such as rail companies that own large quantities of railway networks. In such industries, it is difficult for other firms to enter the market due to high up-front costs; in many cases, an existing firm can block other firms from entering, such as by keeping prices below a sustainable level. AI development might be similar—at the very least, firms within it additional power that allows them to avoid many competitive pressures.

Developing a large model requires paying substantial fixed costs up-front, including expenses for computing power and datasets essential for training. However, once these initial investments are made, the subsequent cost per user for deploying and maintaining these models is considerably lower. Since the average cost per unit decreases as the number of customers increases, it becomes substantially more cost-effective for a single AI company to provide access to many people than for multiple companies to independently develop and maintain similar models [324].

Additionally, only a few companies might have access to enough resources and data to create the best AI models. In the Artificial Intelligence Fundamentals chapter), we discussed how scaling laws demonstrate that improving AI performance has required access to costly resources like high-performance processors and vast, high-quality datasets. High resource requirements for developing advanced AI can limit market entry, stifling competition. Early capability advantages for a leading firm can have positive feedback loops, which could enable them to raise more capital and pull ahead of many competitors. This raises concerns that only a few powerful entities will have access to and benefit from AI technologies.

The use of AI might create or strengthen oligopolies. AI developers are striving to have their models achieve superintelligence: models that are able to carry out a wide range of tasks across various domains better than humans. If someone did manage to create such an AI, they might have a decisive advantage across large swathes of industry, being sufficiently versatile to become an expert in many or every market. It may become difficult or impossible for smaller firms to carve out niche market spaces. The advanced capabilities of general AI may outpace specialized models in diverse domains, making it more difficult for new entrants to gain market share. Large firms equipped with powerful AI systems could wield an enormous amount of power, potentially leading to less competition, higher prices, and slower innovation, hurting both labor and product markets.

Externalities

Externalities are consequences of economic activity that impact unrelated third parties. An externality is a side effect of an economic activity, impacting individuals or groups who are not directly involved in that activity. Because of externalities, market prices for goods or services may not fully reflect the costs that third parties, who are neither the consumers nor the producers of that market, bear as a result of the economic activity.

A classic example of a negative externality—a harm to a third party—is pollution. Consider the Sriracha factory in Irwindale, California, where jalapeño peppers are ground and roasted. Residents of Irwindale claimed that odors from the factory caused lung and eye irritation and created an overall unpleasant smell in the town. The factory, by producing these odors, was imposing a negative externality upon the town’s residents. However, since the townspeople received no automatic compensation for this inconvenience, this was not reflected in the price of Sriracha.

We can resolve externalities with litigation, property rights, and taxation. In 2013, locals sued the Sriracha factory: this legal action led the factory to install new filters to reduce pollution. Litigation can be an effective tool to resolve externalities by forcing compensation. Economic theory suggests that bargaining over the externality can also create efficient outcomes; for instance, if the property right to the air was understood to belong to the townspeople, and that the factory would have to stop polluting or compensate the townspeople at an acceptable rate for their inconvenience [325].

A third commonly used resolution to externalities is taxation. Smoking cigarettes imposes negative externalities on those near the smoker; governments will thus impose taxes on the sale of tobacco, which both raise revenues for the state to run social programs and discourage smoking by increasing the price of cigarettes to better reflect its true total cost. Determining the most effective method for resolving each externality is a topic of ongoing debate among economists. These debates extend to the externalities of AI on society. Potential policy responses are further discussed in the Governance chapter.

The development and deployment of AI systems can lead to negative externalities. AI’s effect on the environment has received a lot of attention [e.g. 326]. Training advanced AI models requires vast computational resources, consuming a significant amount of energy and contributing to greenhouse gas emissions. Emissions can lead to climate change, a cost borne by society at large, rather than only by those who pollute—unless the externality is corrected, such as by charging companies for the carbon they emit. Beyond emissions, several other issues such as worker displacement, the potential for misuse or accidents, and the risk of the loss of control of advanced systems present serious externalities as well.

Moral hazards

Moral hazards occur when risks are externalized. Moral hazards are situations where an entity is encouraged to take on risks, knowing that any costs will be borne by another party. Insurance policies are a classic example: people with damage insurance on their phones might handle them less carefully, secure in the knowledge that any repair costs will be absorbed by the insurance company, not them.

The bankruptcy system ensures that no matter how much a company damages society, the biggest risk it faces is its own dissolution, provided it violates no laws. Companies may rationally gamble to impose very large risks on the rest of society, knowing that if those risks ever come back to the company, the worst case is the company going under. The company will never bear the full cost of damage caused to society due to its risk taking. Sometimes, the government may step in even prior to bankruptcy. For example, leading American banks took on large risks in the lead up to the 2008 financial crisis, but many of them were considered "too big to fail", leading to an expectation that the government would bail them out in time of need [327]. These dynamics ultimately contributed to the Great Recession.

Developing advanced AIs is a moral hazard. In the first chapter, we outlined severe risks to society from advanced AIs. However, while the potential costs to society are immense, the maximum financial downside to a tech company developing these AIs is filing for bankruptcy.

Consider the following, admittedly extreme, scenario. Suppose that a company is on the cusp of inventing an AI system that would boost its profits by a thousand-fold, making every employee a thousand times richer. However, the company estimates that their invention comes with a 0.1% chance of a catastrophic accident leading to large-scale loss of life. In the likely case, the average person in the economy would see some benefits due to increased productivity in the economy, and possibly from wealth redistribution. Still, most people view this gamble as irrational, preferring not to risk catastrophe for modest economic improvements. On the other hand, the company may see this as a worthwhile gamble, as it would make each employee considerably richer.

Risk internalization encourages safer behavior. In the above examples of moral hazards, companies take risks that would more greatly affect external parties than themselves. The converse of this is risk internalization, where risks are primarily borne by the party that takes them. Risk internalization compels the risk-taker to exercise caution, knowing that they would directly suffer the consequences of reckless behavior. If AI companies bear the risk of their actions, they would be more incentivized to invest in safety research, take measures to prevent malicious use, and be generally disincentivized from creating potentially dangerous systems.

6.4.3 Inequality

Most of the world exhibits high levels of inequality. In economics, inequality refers to the uneven distribution of economic resources, including income and

wealth. The *Gini coefficient* is a commonly used statistical measure of the distribution of income within a country. It is a number between 0 and 1, where 0 represents perfect equality (everyone has the same income or wealth), and 1 signifies maximum inequality (one person has all the income, and everyone else has none). Looking at Gini coefficients, 71% of the world's population lives in countries with increasing inequality over the last thirty years [328].

Inequality in the United States is particularly striking. Figure 6.3 shows that the Gini coefficient in the US has trended significantly upwards from 1969 to 2019. Over 50 years, the pre-tax Gini coefficient has increased by 30%, while the post-tax Gini coefficient has risen by 25%, suggesting that despite redistributive taxation policies, the US income gap has widened substantially. (For reference, this change in Gini coefficient is the same size as moving from Canada to Saudi Arabia today. [328]) This increase in the Gini coefficient is evidence of a growing inequality crisis. An associated fall in social mobility—the ability of an individual to move from the bottom income bracket to the top—cements inequalities over generations.

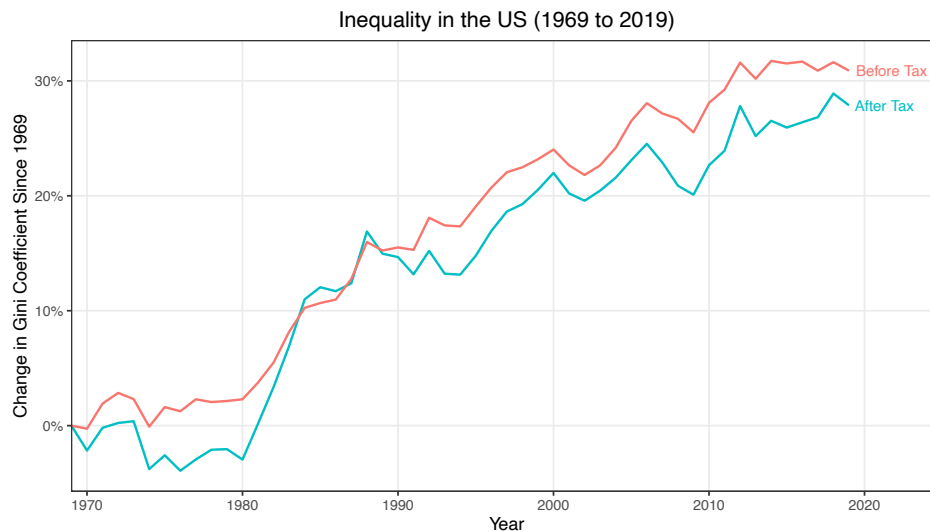


FIGURE 6.3. Inequality in the US (as measured by the Gini coefficient) has risen dramatically over the last five decades, even adjusting for taxation [329].

The distribution of gains from growth is highly unequal. Nearly all the wealth gains over the past five decades have been captured by the top 1% of income earners, while average inflation-adjusted wages have barely increased [330]. A RAND Corporation working paper estimated how the US income distribution would look today if inequality was at the same level as in 1975—the results are in Table 6.1. Suppose my annual income is \$15,000 today. If inequality was at the same level as in 1975, I would be paid an extra \$5,000. Someone else earning \$65,000 today would instead have been paid \$100,000 had inequality held constant! We can see in the table below that these increases in inequality have had massive effects on individual incomes for everyone outside the top 1%.

TABLE 6.1. Real and counterfactual income distributions for all adults with income, in 2018 USD [331].

Percentile	Actual Income in 1975	Actual Income in 2018	Income in 2018 if Inequality Had Stayed Constant
25th %	\$9,000	\$15,000	\$20,000
Median	\$26,000	\$36,000	\$57,000
75th %	\$46,000	\$65,000	\$100,000
90th %	\$65,000	\$112,000	\$142,000
95th %	\$80,000	\$164,000	\$174,000
99th %	\$162,000	\$491,000	\$353,000
Top 1% Mean	\$252,000	\$1,160,000	\$549,000

Inequality carries serious implications. Beyond obvious problems like an inability to access essentials and maintain a basic standard of living, those worst off in an unequal society face additional problems in domains like health. A widening wealth gap often corresponds with a health gap, where those with fewer resources have poorer health outcomes due to less access to quality healthcare, lower quality nutrition, and higher stress levels. For instance, life expectancy often varies dramatically based on income in unequal societies [332].

Everyone, not just the poorest, suffers in an unequal society. One of the most robust findings about inequality is that unequal societies have higher levels of crime [333]. When inequality is high, so too are levels of social tension, dissatisfaction, and shame, which can contribute to higher crime rates. This can lead to a cycle where the fear of crime drives further inequality, as wealthier individuals and neighborhoods invest in measures that segregate them further from the rest of society, further increasing inequality and crime rates. A more detailed discussion of this relative deprivation can be found in the section on Cooperation and Conflict. Inequality is also related to other signs of societal sickness: worse physical and mental health, increased drug use, and higher rates of incarceration. Strikingly, inequality is a strong predictor of political instability and violence as well. While it may seem like those with more wealth are insulated from the negative effects of inequality, they suffer indirect consequences as well.

The causes of inequality are disputed. Renowned economist Thomas Piketty, in his book “Capital in the 21st Century” posits that inequality is a consequence of the difference between the interest rate investments in capital receive (r) and the rate of economic growth (g) [337]. Piketty’s key unorthodox claim is that capital is a “gross substitute” for labor. On this view, as capital owners generate more wealth and capital, they are increasingly able to use capital to replace labor—imagine this as robots outcompeting human workers. Piketty argues that if the interest rate, which is the rate at which capital (or “robots”) can self-replicate, is greater than the overall rate of economic growth, and thus faster than labor’s wage growth, then capital owners can become substantially and continuously richer than workers. This contrasts with the standard view that capital is a “gross complement” to labor, with

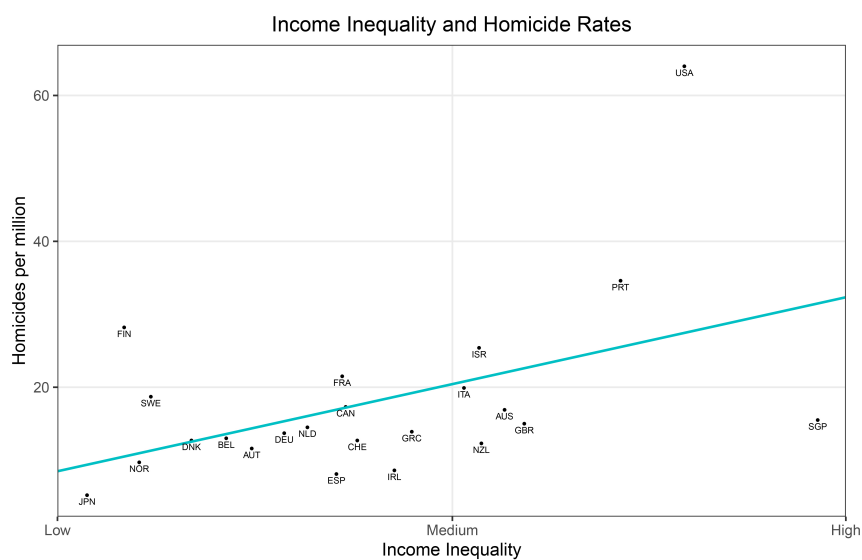


FIGURE 6.4. Countries with higher income inequality tend to have higher homicide rates [334, 335].

both labour and capital needed to produce goods. According to the standard view, increases in capital lead to higher labor productivity, which makes workers more efficient and valuable. Increased productivity raises wages; thus, increases in capital benefit both workers and capital owners.

If AI serves as a gross substitute for labour, investment in AIs, with an effective interest rate (r) higher than the overall growth rate (g), will permit capital owners to con-

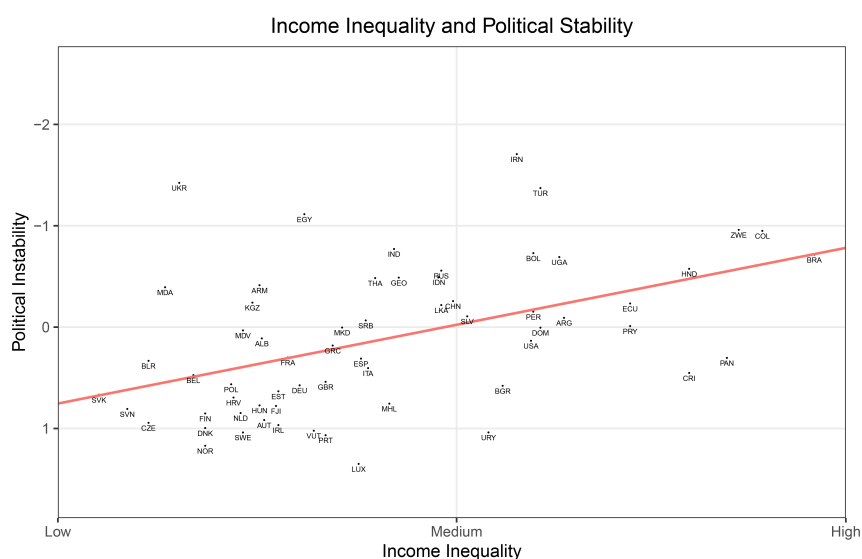


FIGURE 6.5. Countries with higher income inequality tend to have higher rates of political instability [334, 336].

tinue accumulating capital, outcompeting workers and increasing inequality. While on the standard view, this would be a fundamentally new phenomenon, Piketty would argue that this is the exacerbation of a centuries-old trend. Such a scenario would contribute to growing inequality and negatively impact the livelihoods of workers. Issues of automation through AI and its broader societal consequences are discussed further in the Governance chapter.

6.4.4 Growth

Growth is widely considered essential to a healthy society. Some have claimed that societies should place less importance on economic growth, whether to reduce environment impacts or for other reasons. However, looking at the world over the last 200 years provides strong evidence that economic growth can lead to vast improvements in human welfare in the domains of health, education, and more.

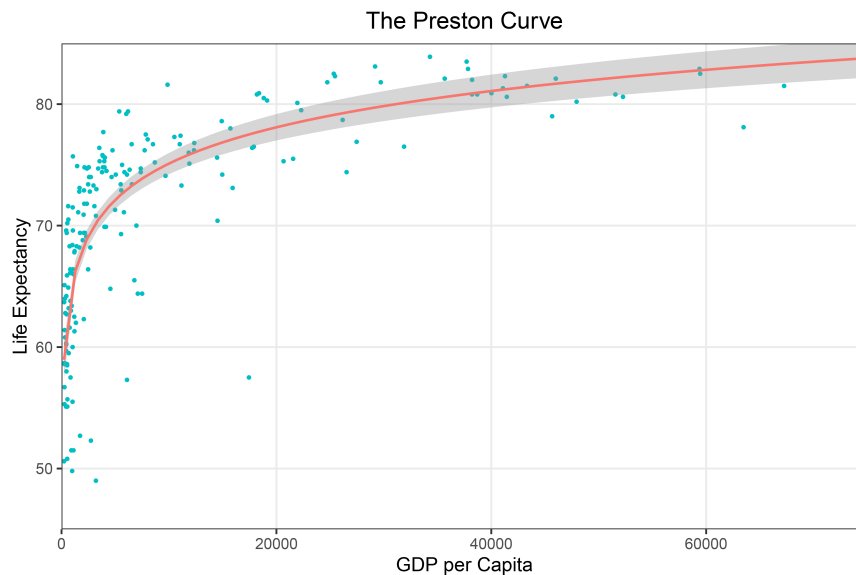


FIGURE 6.6. Increases in GDP per capita strongly correlate with increases in life expectancy [338].

We have strong reasons for encouraging economic growth. The Preston curve is compelling evidence of a positive correlation between a country's gross domestic product (GDP) per capita—a measure of the total production of goods and services—and health outcomes. This relationship illustrates that countries with higher GDP per capita have better health outcomes. The shape of the relationship indicates that poorer countries, in particular, stand to benefit immensely from improvements in GDP. The positive correlation also holds across time: average global life expectancy was just below 40 years at the start of the 20th century, whereas today, with a much higher global GDP per capita, the average person expects to live for 70 years. Nobel laureate Amartya Sen suggests that one pathway through

which growth improves health is by reducing poverty and increasing investments in healthcare [339].

Growth makes it easier for societies to support a wide range of values.

The benefits of economic growth extend beyond physical health. There is a strong association between economic prosperity and enhancements in freedom and education [340]. Prosperous societies can afford stronger institutions to safeguard democratic freedoms and human rights. As societies become wealthier, more resources can be allocated to support cultural institutions, artists, and creative endeavors. Economic growth can enable *Pareto improvements*: changes that improve people's lives without leaving anyone worse off, like growing a pie to give everyone bigger slices. Instead of debating whether to spend limited resources on a new hospital, school, or cultural center, encouraging economic growth can allow us to build all three. The increased wealth effectively allows us to accommodate multiple values, protecting *value pluralism*.

6.4.5 Beyond Economic Models

However, in measuring social wellbeing, we must recognize some shortcomings of traditional economic metrics. Here, we will discuss the disconnect between economic output and social value, why relying on economic models of welfare economics can be inadequate to describe human goals, and how more holistic measures of happiness and economic prosperity may give us a clearer sense of true social wellbeing.

Economic Output and Gross Domestic Product

Economic indicators measure what we can quantify, not necessarily what we care about.

Indicators like Gross Domestic Product (GDP) measure the monetary value of final goods and services—that is, those that are bought by the final user—produced in a country in a given period of time. When economists discuss growth, they are typically referring to increases in GDP. Measures of productive output like GDP are useful in gauging a country's economic health, but they fail to capture the value of socially significant activities that aren't priced in the market.

Some socially important tasks are not captured by GDP.

Many essential roles in society, such as parenthood, community service, and early education, are crucial to the wellbeing of individuals and communities but are often undervalued or entirely overlooked in GDP calculations [341]. While their effects might be captured—education, for instance, will increase productivity, which increases GDP—the initial activity does not count towards total output. The reason is simple: GDP only accounts for activities that have a market price. Consequently, efforts expended in these socially important tasks, despite their high intrinsic value, are not reflected in the GDP figures.

Technologies that make our lives better may not be measured either.

Technological advancements and their value often fail to be reflected adequately in

GDP figures. For instance, numerous open-source projects such as Wikipedia provide knowledge to internet users worldwide at no cost. However, because there's no direct monetary transaction involved, the immense value they offer isn't represented in GDP. The same applies to user-generated content on platforms like YouTube, where the main contribution to GDP is through advertisement revenue because most creators aren't compensated directly for the value they create. The value viewers derive from such platforms vastly outstrips the revenue generated from ads or sales on these platforms, but this is not reflected in GDP.

There might be a similar disconnect between GDP and the social value of AI. As artificial intelligence systems become more integrated into our daily lives, the disconnect between GDP and social value might become more pronounced. For example, an AI system that provides free education resources may significantly improve learning outcomes for millions, but its direct contribution would be largely invisible in GDP terms. Conversely, an AI may substantially increase GDP by facilitating high-frequency trading without doing much to increase social wellbeing. This growing chasm between economic metrics and real value could lead to policy decisions that fail to harness the full potential of AI or inadvertently hamper its beneficial applications.

The proxy-purpose distinction is especially important for AI. Imagine a future where an AI system is tasked with maximizing GDP, often seen as a proxy for wellbeing. The system could potentially achieve this goal by promoting resource-intensive industries or fostering a work culture that prioritizes productivity over wellbeing. In such a scenario, the GDP might increase, but at the cost of essential considerations like environmental sustainability or human happiness. Therefore, relying solely on economic indicators could lead to decisions that, while effective in the short term, might harm society in the long run.

Understanding the limitations of economic measurements is an important step towards a safer use of AI. It helps us question what we should optimize. The aim of economic policy should not just be to maximize economic output but also to promote overall social wellbeing. AI systems may thus need to take into account multiple factors, like equality, sustainability, and personal fulfillment, rather than only economic indicators.

Models of Welfare Economics

The most basic form of welfare economics maximizes social surplus. Social surplus is a measure of the total value created by a market: it is the sum of the consumer surplus and the producer surplus. The consumer surplus is the difference between the maximum price a consumer is willing to pay and the actual market price. Conversely, the producer surplus is the difference between the actual market price and the minimum price a producer is willing to accept for a product or service. By maximizing the total surplus, welfare economics seeks to maximize the social value created by a market.

As an example, imagine a scenario where consumers are willing to pay up to \$20 for a book, but the market price is only \$15. Here, the consumer surplus is \$5. Similarly, if a producer would be willing to sell the book for a minimum of \$10, the producer surplus is \$5. The social surplus, and hence the social value in this market, is \$10: \$5 consumer surplus plus \$5 producer surplus.

We should be wary of generalizing from simple models. The core model of welfare economics has its limitations. Notably, welfare economics is concerned with the maximization of surplus, but is indifferent to its distribution. This might not align with common notions of fairness and equality. For example, an AI optimized to maximize profits might model consumers well enough to enable perfect price discrimination: allowing firms to sell each good at exactly a consumer's maximum willingness to pay, converting all the consumer surplus into producer surplus, but leaving the sum total of "social surplus" unchanged.

However, social surplus is not the only thing we care about. Since people tend to derive less utility from a marginal dollar when they are richer, we care about how rich consumers and producers are to begin with. If, as is usually the case, consumers are poorer than the owners of a company, then transferring \$5 (of surplus) from consumers to producers decreases total utility. The real gains for the firm may be small compared to the losses for the consumers.

As AI systems become more integral to our economies, we must be mindful of these complexities. A narrow focus on maximizing economic surplus could lead us to deploy AIs which, while efficient in a purely economic sense, might have harmful consequences for society.

Happiness in Economics

There is a gap between human happiness and material prosperity. Most people would agree that the goal of social sciences should not be to just increase material wealth. A more meaningful aim would be to enhance overall wellbeing or happiness. However, defining and measuring happiness can be challenging. Whether happiness is correlated with material wealth remains an ongoing research question; other aspects of life like physical and mental health, job satisfaction, social connections, and a sense of purpose seem important as well.

Debates about the Easterlin paradox highlight the complexity of understanding happiness. While wealthier people and countries are generally happier than their less affluent counterparts, long-term economic growth does not always correlate with long-term increases in happiness: this is the Easterlin paradox. Several studies have tried to explore the relationship between happiness and economic growth [342]. While some findings suggest a correlation, others don't: our understanding of the happiness-growth relationship is still evolving.

We do, however, have strong evidence that inequality is harmful. People often evaluate their wellbeing in relation to others; so, when wealth distribution is unequal, people are dissatisfied and unhappy. For instance, the recent rise in inequality may explain

why there has been no significant increase in happiness in the US over the last few decades despite an approximately tenfold increase in real GDP and a fourfold increase in real GDP per capita.

More holistic economic measurements can get closer to capturing what we value. Due to the disconnect between economic prosperity and true wellbeing, many economists also use broader metrics. These measures aim to capture wellbeing more comprehensively, rather than solely focussing on economic growth. One such measure is the Human Development Index (HDI). The HDI comprises a nation's average life expectancy, education level, and Gross National Income (GNI) per capita (which is similar to GDP). Notably, the UN uses the logarithm of GNI per capita in the HDI calculation, which accounts for the diminishing returns of wealth: the idea that each additional dollar earned contributes less to a person's happiness than the one before it. In general, economists consider a "report card" of indicators to assess a nation's wellbeing, rather than just depending on a single measure. By capturing various aspects of wellbeing, this approach could provide a more holistic and accurate representation of a nation's quality of life.

Summary. Traditional economic measures and models are insufficient for measuring and modeling what we care about. There is a disconnect between what we measure and what we value; for instance, GDP fails to account for essential unpaid labor and overvalues the production of goods and services that add little to social wellbeing. While economic models are useful, we must avoid relying too much on theoretically appealing models and examine the matter of social wellbeing with a more holistic lens.

Conclusions About the Economic Engine

We should be wary of using AI to increase metrics that are only proxies for wellbeing. Our current economic system strongly incentivises the deployment of AI systems that optimize for economic growth which, while often a worthy goal, may not capture essential aspects of societal health such as equality or sustainability (due to unequal distribution of gains, externalities, and other market failures). While economic objectives like GDP growth are quantifiable and easy to pursue, they may not truly reflect what makes a society happy and healthy.

It is risky to let the goals of AI systems be determined entirely by economic incentives. AI systems created by for-profit businesses are designed to maximize shareholder profits, not societal wellbeing. If we let the economy decide what AIs do by letting largely unregulated markets create AIs, we may end up with an increase in inequality and exacerbation of market failures. Using money as a proxy for social value might seem practical, but it can distort societal priorities; for instance, this system implies that the preferences of wealthier individuals hold more weight since they are willing and able to spend more.

The examples in this section demonstrate a divergence between economic incentives and other important societal goals and should serve as a cautionary note. In the next

section, we consider an alternative view: a framework that puts happiness front and center. Perhaps if we can direct AI systems to focus directly on promoting human happiness, we might aim to overcome the human biases and limitations that stop us from pursuing our happiness and enable the system to make decisions that have a positive impact on overall wellbeing.

6.5 WELLBEING

In the next few sections, we will explore how AIs can be used to increase human wellbeing. We start by asking: what is wellbeing?

Wellbeing can be defined as how well a person's life is going for them. It is commonly considered to be intrinsically good, and some think of wellbeing as the ultimate good. Utilitarianism, for instance, holds some form of wellbeing as the sole moral good.

There are different accounts of precisely what wellbeing is and how we can evaluate it. Generally, a person's wellbeing seems to depend on the extent to which that person is happy, healthy, and fulfilled. Three common accounts of wellbeing characterize it as the net balance of pleasure over pain, a collection of objective goods, or preference satisfaction. Each account is detailed below.

6.5.1 Wellbeing as the Net Balance of Pleasure over Pain

Some philosophers, known as *hedonists*, argue that wellbeing is the achievement of the greatest balance of pleasure and happiness over pain and suffering. (For simplicity we do not distinguish, in this chapter, between “pleasure” and “happiness” or between “pain” and “suffering”). All else equal, individuals who experience more pleasure have higher wellbeing and individuals who experience more pain have lower wellbeing.

According to hedonism, pleasure is the only intrinsic good. Goods like health, knowledge, and love are instrumentally valuable. That is, they are only good insofar as they lead to pleasure. It may feel as though other activities are intrinsically valuable. For instance, someone who loves literature may feel that studying classic works is valuable for its own sake. Yet, if literature lovers were confronted with proof that reading the classics makes them less happy than they otherwise would be, they might no longer value studying literature. Hedonists believe that when we think we value certain activities, we actually value the pleasure they bring us, not the activities themselves.

Hedonism is a relatively clear and intuitive account of wellbeing. It seems to apply equally to everyone. That is, while we all may have different preferences and desires, pleasure seems to be universally valued. However, some philosophers argue that hedonism is an incomplete account of wellbeing. They argue there may be other factors that influence wellbeing, such as the pursuit of knowledge.

6.5.2 Wellbeing as a Collection of Objective Goods

Others believe that wellbeing is the achievement of an objective set of “goods” or “values.” These goods are considered necessary for living a good life regardless of how pleasurable someone’s life is.

There is disagreement about which goods are necessary for wellbeing. Commonly proposed goods include pleasure, happiness, health, relationships, knowledge, and more. Objective goods theorists consider these values to be important for wellbeing independently of individual beliefs and preferences.

One objection to the objective goods theory is that it is elitist. The objective goods theory claims that some things are good for people even if they derive no pleasure or satisfaction from them. This claim might seem objectionably paternalistic; for instance, it seems condescending to claim that someone with little regard for aesthetic appreciation is thereby leading a deficient life. In response, objective goods theorists might claim that these additional goods do benefit people, but only if those people do in fact enjoy them.

There is no uncontroversial way to determine which goods are important for living a good life. However, this uncertainty is not a unique problem for objective goods theory. It can be difficult for hedonists to explain why happiness is the only value that is important for wellbeing, for instance. In the following sections we focus primarily on other interpretations of wellbeing and do not have space to discuss objective goods theory in depth, particularly given that there are many ways it can be specified.

6.5.3 Wellbeing as Preference Satisfaction

Some philosophers claim that what really matters for wellbeing is that our preferences are satisfied, even if satisfying preferences does not always lead to pleasurable experiences. One difficulty for preference-based theories is that there are different kinds of preferences, and it’s unclear which ones matter. Preferences can be split into three categories: stated preferences, revealed preferences, and idealized preferences. If someone expresses a preference for eating healthy but never does, then their stated preference (eating healthy) diverges from their revealed preference (eating unhealthy). Suppose they would choose to eat healthy if fully informed of the costs and benefits: their idealized preferences, then, would be to eat healthy. Each of these categories can be informative in different contexts: we explore their relevance in the next section.

A Note on Wellbeing in Social Science

Philosophers continue to debate what wellbeing is, or what it means to live a good life. However, over the years, researchers have developed ways to approximate wellbeing for practical purposes. Psychologists, economists, philanthropists, policy makers, and other professionals need—at least—a working definition of wellbeing in order to study, measure, and promote it. Here, we

describe some common views of wellbeing and illustrate how they are used in social science.

Preference satisfaction. Preference theorists view wellbeing as fulfilling desires or satisfying preferences, even if doing so does not always induce pleasure. Nonetheless, it remains an open question whether all desires are tied to wellbeing or just certain kinds, like higher-order or informed desires.

Economics. *Standard economics* often uses preference satisfaction theories to study wellbeing. Revealed preferences can be observed by studying the decisions people make. If people desire higher incomes, for example, economists can promote wellbeing by researching the impact of different economic policies on gross domestic product (*GDP*).

Social Surveys and Psychology. Traditionally, psychological surveys evaluate wellbeing in terms of *life satisfaction*. Life satisfaction is a measure of people's stated preferences—preferences or thoughts that individuals outwardly express—regarding how their lives are going for them. Life satisfaction surveys typically focus on tangible characteristics of people's lives, such as financial security, health, and personal achievements. They are well suited to understanding the effects of concrete economic factors, such as income and education, on an individual's psychological wellbeing. For example, to promote wellbeing, psychologists might research the effects of access to education on one's ability to achieve the goals they set for themselves.

Hedonism. Under hedonist theories of wellbeing, an individual's wellbeing is determined by their mental states. In particular, the balance of positive mental states (like pleasure or happiness) over negative mental states (like pain or suffering).

Economics. *Welfare economics* uses hedonism to evaluate the wellbeing of populations. To estimate gross national happiness (*GNH*) – an indicator of national welfare – it considers the effects of several factors from psychological wellbeing to ecological diversity and resilience on individuals' mental states. Welfare economists might prefer this framework because it is more holistic – it evaluates both material and non-material aspects of everyday life as they contribute to national welfare.

Social Surveys and Psychology. Many psychologists also use hedonist theories to understand and promote wellbeing. They may work to identify the emotional correlates of happiness through surveys that measure people's stated emotions – unlike life satisfaction surveys, these surveys do not reveal mental states. They reveal emotions that people *remember*, not emotions they currently *experience*. Researchers continue to look for ways to directly observe emotions as they are experienced. For example, some studies use cell phone apps to periodically prompt participants to record their current emotions. Such research tactics may provide us with more precise measures of individu-

als' overall *happiness* by evaluating the emotional responses to their everyday experiences in near real-time.

Objective goods. Under objective goods theories, wellbeing is determined by a certain number of observable factors, independent of individuals' preferences or experiences. There are multiple theories about what those factors may be. One of the most widely supported theories is *human flourishing*. Under this view, wellbeing is more than just the balance of pleasure over suffering, or the fulfillment of one's preferences – “the good life” should be full, virtuous, and meaningful, encapsulating psychological, hedonistic, and social wellbeing all at once.

Economics. In economics, the *capabilities approach* defines wellbeing as having access to a set of capabilities that allow one to live the kind of life they value. It emphasizes two core ideas: *functionings* and *capabilities*. Functionings include basic and complex human needs, ranging from good health to meaningful relationships. Capabilities refer to the ability people have to choose and achieve the functionings they value – they may include being able to move freely or participate in the political process. This approach has significantly influenced human development indicators, such as the Human Development Index (*HDI*) – it allows developmental economists to measure and compare wellbeing across different populations while also evaluating the effectiveness of public policies.

Psychology. Positive psychologists do not collapse wellbeing into one dimension, rather, they argue for a *psychologically rich life* – one that is happy, meaningful, and engaging. Some psychologists use *PERMA theory* to evaluate psychological richness, which considers five categories essential to human flourishing: 1) experience of *positive emotions*, 2) *engagement* with one's interests, 3) maintenance of personal, professional, and social *relationships*, 4) the search for *meaning* or purpose, and 5) *accomplishments*, or the pursuit of one's goals. This framework is particularly useful in evaluating wellbeing because it is universal - it can be applied cross-culturally - and practical - it can guide interventions aiming to improve emotional wellbeing, social relationships, or activities that provide a sense of meaning or accomplishment.

While we don't have a complete understanding of the nature of wellbeing, we can use these theories as useful research tools. They can help us to (a) understand how different factors contribute to wellbeing and (b) evaluate the effectiveness of policies and other interventions aimed at improving wellbeing.

6.5.4 Applying the Theories of Wellbeing

While people disagree about which account of wellbeing is correct, most people agree that wellbeing is an important moral consideration. All else equal, taking actions that promote wellbeing is generally considered morally superior to taking actions that

reduce wellbeing. However, the three theories of wellbeing have different practical implications. In the future, we are likely to interact with AI chatbots in a variety of ways; in particular, we might have close personal interactions with them that influence our decision-making. Different theories of wellbeing would suggest different goals for these AIs.

Chatbots could prioritize pleasure. The hedonistic view suggests that wellbeing is primarily about experiencing pleasure and avoiding pain. This theory might recommend that AIs should be providing users with entertaining content that brings them pleasure or encouraging them to make decisions that maximize their balance of pain over pleasure over the long run. A common criticism is that this trades off with other goods considered valuable like friendship and knowledge. While this is sometimes true, these goals can also align. Providing content that is psychologically rich and supports users' personal growth can contribute to a more fulfilling and meaningful life full of genuinely pleasurable experiences.

Chatbots could prioritize preference satisfaction. The preference view suggests that wellbeing is about having preferences satisfied. Depending on which preferences were considered important, this theory would suggest different priorities for a hedonistic chatbot. Consider revealed preferences. One proxy for revealed preferences is user engagement. By continuing to engage with a chatbot, users are expressing their preference for seeing more of what they are getting, and increasing preference satisfaction might imply continuing to behave in certain ways. It is important to be careful of this equivalence, though: prioritizing user engagement can lead to results like chatbots that engage people through unsavory means. Like engaging humans, chatbots could try to addict users by showing them streams of ephemeral content, creating an air of mystery and uncertainty, or act distant after being friendly to create a desire for continued friendliness that it can then satisfy. Such AIs might maximize engagement, but this may not be good for people's wellbeing—even if they use the platform for more hours.

Chatbots could promote objective goods. The objective goods account suggests that wellbeing is about promoting goods such as achievement, relationships, knowledge, beauty, happiness, and the like. An AI chatbot might aim to enhance users' lives by encouraging them to complete projects, develop their rational capacities, and facilitate learning. The goal would be to make users more virtuous and encourage them to strive for the best version of themselves. This aligns with Aristotle's theory of friendship, which emphasizes the pursuit of shared virtues and mutual growth, suggesting that such AIs might have meaningful friendships with humans.

We might want to promote the welfare of AIs. In the future, we might also come to view AIs as able to have wellbeing. This might depend on our understanding of wellbeing. An AI might have preferences but not experience pleasure, which would mean it could have wellbeing according to preference satisfaction theorists but not hedonists. Future AIs may have wellbeing according to all three accounts of wellbeing. This would potentially require that we dramatically reassess our relationship with AIs. This question is further discussed in the Happiness section in this chapter.

In the following sections, we will focus on the different conceptions of wellbeing presented here, and explore what each theory implies about how we should embed ethics into AIs.

A Note on Measuring Wellbeing

While the philosophical foundations of wellbeing are not settled, quantitative research fields like public health and economics require the use of metrics in order to evaluate, track, or compare the subject of study. Researchers use many different metrics to measure wellbeing, but the most common are HALYs and WELBYs.

Health-adjusted life years (HALYs). A very common unit for measuring wellbeing is the health-adjusted life year, or HALY. HALYs account for two factors: (1) the number of years lived by an individual, group, or population (also called “life years”), and (2) the health of those life years. Two common types of HALYs are QALYs and DALYs.

Quality-adjusted life years (QALYs). QALYs measure the number of years lived, adjusted according to health. One year of life in perfect health is equivalent to 1 QALY. One year of a less healthy life is worth between 0 and 1 QALYs. The value of a life year depends on how severely the life is impacted by health problems. For example, a year of life with asthma might be worth 0.9 QALYs, while a year of life with a missing limb might be worth about 0.7 QALYs.

Disability-adjusted life years (DALYs). While QALYs measure years of life as impacted by health, DALYs measure years of life lost, accounting for the impact of health. Whereas 1 QALY is equivalent to a year in perfect health, 1 DALY is equivalent to the loss of a year in perfect health. A year of life with asthma might be worth 0.1 DALYs, while a year of life with a missing limb might be worth 0.3 DALYs.

Note that increases in wellbeing are indicated by higher numbers of QALYs but lower numbers of DALYs.

Using HALYs to measure wellbeing has some limitations. First, the extent to which different illnesses or injuries affect overall human health is not clear. Losing a limb probably has a larger health impact than getting asthma, but researchers must rely on subjective judgements to assign precise values to each problem. Second—and perhaps more importantly—HALYs measure the value of a span of life as it is impacted by health alone. In reality, there are many factors that can impact the value of life, like happiness, relationship quality, freedom, and a sense of purpose. Perhaps a more useful measurement of wellbeing would consider the effects of all of these factors, rather than health alone.

Wellbeing-adjusted life years (WELBYs). WELBYs have been developed to measure years of life as impacted by overall wellbeing. One WELBY

is equivalent to one year of life at maximum wellbeing — namely, a life that is going as well as possible. Wellbeing can be assessed using self-reported outcomes like affect, life satisfaction, or degree of flourishing. There may also be some empirical ways to assess wellbeing like cortisol levels, income, or ability.

QALYs, DALYs, and WELBYs provide different approximations of wellbeing that can be used to inform high-level decision-making and policy-setting.

6.6 PREFERENCES

Should we have AIs satisfy people’s preferences? A preference is a tendency to favor one thing over another. Someone might prefer chocolate ice cream over vanilla ice cream, or they might prefer that one party wins the election rather than another. These preferences will influence actions. If someone prefers chocolate ice cream over vanilla, they’re more likely to choose the former. Similarly, if someone prefers one political party over another, they will likely vote accordingly. In this manner, our preferences shape our behavior, guiding us toward certain choices and actions over others. Preference is similar to desire but always comparative. Someone might desire something in itself—a new book, a vacation, or a delicious meal—but a preference always involves a comparison between two or more alternatives.

Overview. In this section, we will consider whether preferences may have an important role to play in creating AIs that behave ethically. In particular, if we want to design an advanced AI system, the preferences of the people affected by its decisions should plausibly help guide its decision-making. In fact, some people (such as preference utilitarians) would say that preferences are all we need. However, even if we don’t take this view, we should recognize that preferences are still important.

To use preferences as the basis for increasing social wellbeing, we must somehow combine the conflicting preferences of different people. We’ll come to this later in this chapter, in a section on social welfare functions. Before that, however, we must answer a more basic question: what exactly does it mean to say that someone prefers one thing over another? Moreover, we must decide why we think that satisfying someone’s preferences is good for them and whether all kinds of preferences are equally valuable. This section considers three different types of preferences that could all potentially play a role in decision-making by AI systems: revealed preferences, stated preferences, and idealized preferences.

6.6.1 Revealed Preferences

Preferences can be inferred from behavior. One set of techniques for getting AI systems to behave as we want—inverse reinforcement learning—is to have them deduce *revealed preferences* from our behavior. We say that someone has a revealed preference for X over Y if they choose X when Y is also available. In this way,

preference is revealed through choice. Consider, for example, someone deciding what to have for dinner at a restaurant. They're given a menu, a list of various dishes they could order. The selection they make from the menu is seen as a demonstration of their preference. If they choose a grilled chicken salad over a steak or a plate of spaghetti, they've just revealed their preference for grilled chicken salad, at least in that specific context and time.

While all theories of preferences agree that there is an important link between preference and choice, the revealed preference account goes one step further and claims that preference simply *is* choice.

Revealed preferences preserve autonomy. One advantage of revealed preferences is that we don't have to guess what someone prefers. We can simply look at what they choose. In this way, revealed preferences can help us avoid paternalism. Paternalism is when leaders use their sovereignty to make decisions for their subjects, limiting their freedom or choices, believing it is for the subjects' own good. However, we may think that typically people are themselves the best judges of what is good for them. If so, then by relying on people's actions to reveal their preferences, we avoid the risk of paternalism.

However, there are problems with revealed preferences. The next few subsections will explore the challenges of misinformation, weakness of will, and manipulation in the context of revealed preferences. We will discuss how misinformation can lead to choices that do not accurately reflect a person's true preferences, and how weakness of will can cause individuals to act against their genuine preferences. Additionally, we will examine the various ways in which preferences can be manipulated, ranging from advertising tactics to extreme cases like cults, and the ethical implications of preference manipulation.

Misinformation

Revealed preferences can sometimes be based on misinformation. If someone buys a used car that turns out to be defective, it doesn't mean they prefer a faulty car. They intended to buy a reliable car, but due to a lack of information or deceit from the seller, they ended up with a substandard one. Similarly, losing at chess doesn't indicate a preference for losing; rather, it's an outcome of facing a stronger player or making mistakes during the game. This means that we cannot always infer someone's preferences from the choices they make. Choice does not reveal a preference between things as they actually are, but between things as the person understands them. Therefore, we can't rely on revealed preferences if they are based on misinformation.

Weakness of Will

Choices can be due to a lack of willpower rather than considered preferences. Consider a smoker who wants to quit. Each time they light a cigarette, they may be acting against their genuine preference to stop smoking, succumbing instead

to the power of addiction. Therefore, it would be erroneous to conclude from their behavior that they think that continuing to smoke would be best for their wellbeing.

Manipulation

Revealed preferences can be manipulated in various ways. Manipulations like persuasive advertising might manipulate people into buying products they don't actually want. Similarly, revealed preferences might be the result of social pressure rather than considered judgment, such as when buying a particular style of clothing. In such cases, the manipulation may not be especially malicious. At the other extreme, however, cults may brainwash their members into preferring things they would not otherwise want, even committing suicide. In the context of decision-making by advanced AI systems, manipulation is a serious risk. An AI advisor system could attempt to influence our decision-making for a variety of reasons—perhaps its designer wants to promote specific products—by presenting options in some particular way; for instance, by presenting a list of options that looks exhaustive and excluding something it doesn't want us to consider.

If preference satisfaction is important, perhaps manipulation is acceptable. In at least some of these cases, it seems clear that preference manipulation is bad. However, it may be less clear exactly why it is bad. A natural answer is to say that people might be manipulated into preferring things that are bad for them. Someone who is manipulated by advertising into preferring junk food might thereby suffer negative health consequences. However, if we think that wellbeing simply consists in preference satisfaction, it doesn't make sense to say that we might prefer what is bad for us. On this account, having one's preferences satisfied is by definition good, regardless of whether those preferences have been manipulated. This might lead us to think that what matters is not (or at least not only) preference satisfaction, but happiness or enjoyment. We'll discuss this in the section on happiness.

Disliking manipulation suggests that wellbeing requires autonomy. On the other hand, some may find that manipulation is bad even if the person is manipulated into doing something that is good for them. For example, suppose a doctor lies to her patient, telling him that unless he loses weight, he will likely die soon. As a result, the patient becomes greatly motivated to lose weight and successfully does so. This provides a range of health benefits, even if his doctor never had any reason to believe he would have died otherwise. If we think manipulation is still bad, lack of enjoyment can't be the whole story. This suggests that we object to manipulation in part because it violates autonomy. We might then think that autonomy—the ability to decide important matters for oneself, without coercion—is objectively valuable regardless of what the agent prefers.

Inverse Reinforcement Learning

Inverse Reinforcement Learning (IRL) relies on revealed preferences. IRL is a powerful technique in the field of machine learning, which focuses on extracting an agent's objectives or preferences by observing its behaviors. In more technical

terms, IRL is about reverse engineering the reward function—an internal ranking system that an agent uses to assess the value of different outcomes—that the agent appears to be optimizing, given a set of its actions and a model of the environment. This technique can help ensure the alignment of AI system’s behaviors with human values and preferences. However, leveraging revealed preferences or observable choices of humans to train AI systems using IRL poses significant challenges pertaining to AI safety.

Using revealed preferences as a training mechanism for IRL can be risky.

Reconsider the chess example: losing a game does not mean that we prefer to lose. This interpretation could be a misrepresentation of the player’s true preferences, potentially leading to undesirable outcomes. Furthermore, extending observed preferences to unfamiliar situations poses another hurdle. I may prefer to eat ice cream for dessert, but that doesn’t mean I prefer to eat it for every meal. Similarly, I may prefer to wear comfortable shoes for hiking, but that doesn’t mean I want to wear them to a formal event. An AI system could inaccurately extrapolate preferences from limited or context-specific data and misapply these to other scenarios. Therefore, while revealed preferences can offer significant insights for training AI, it is vital to understand their limitations to safeguard the safety and efficiency of AI systems.

Summary. Revealed preferences can be a powerful tool, as they allow an individual’s actions to speak for themselves, reducing the risk of paternalistic intervention. However, revealed preferences have inherent shortcomings such as susceptibility to misinformation and manipulation, which can mislead an AI system. This emphasizes the caution needed in relying solely on revealed preferences for AI training. It underscores the importance of supplementing revealed preferences with other methods to ensure a more comprehensive and accurate understanding of a user’s true preferences.

6.6.2 Stated Preferences

Preferences can be straightforwardly queried. Another set of techniques for getting AI systems to behave as we want—human supervision and feedback—rely on people to state their preferences. A person’s **stated preferences** are the preferences they would report if asked. For example, someone might ask a friend which movie they want to see. Similarly, opinion polls might ask people which party they would vote for. In both cases, we rely on what people say as opposed to what they do, as was the case with revealed preferences.

Stated preferences overcome some of the difficulties with revealed preferences. For example, stated preferences are less likely to be subject to weakness of will: when we are further removed from the situation, we are less inclined to fall for temptations. Therefore, stated preferences are more likely to reflect our stable, long-term interests.

Stated preferences are still imperfect. Stated preferences do not overcome all difficulties with revealed preferences. Stated preferences can still be manipulated. Further, individuals might state preferences they believe to be socially acceptable

or admirable rather than what they truly prefer, particularly when the topics are sensitive or controversial. Someone might overstate their commitment to recycling in a survey, for instance, due to societal pressure and norms around environmental responsibility.

Preference Accounting

One set of problems with stated preferences concerns which types of preferences should be satisfied.

First, someone might never know their preference was fulfilled. Suppose someone is on a trip far away. On a bus journey, they exchange a few glances with a stranger whom they'll never meet again. Nevertheless they form the preference that the stranger's life goes well. Should this preference be taken into account? By assumption, they will never be in a position to know whether the preference has been satisfied or not. Therefore, they will never experience any of the enjoyment associated with having their preference satisfied.

Second, we may or may not care about the preferences of the dead. Suppose someone in the 18th century wanted to be famous long after their death. Should such preferences count? Do they give us reason to promote that person's fame today? As in the previous example, the satisfaction of such preferences can't contribute any enjoyment to the person's life. Could it be that what we really care about is enjoyment or happiness, and that preferences are a useful but imperfect guide toward what we enjoy? We will return to this in the section on happiness.

Third, preferences can be about others' preferences being fulfilled. Suppose a parent prefers that their children's preferences are satisfied. Should this preference count, in addition to their children's preferences themselves? If we say yes, it follows that it is more important to satisfy the preferences of those who have many people who care for them than of those who do not. One might think that this is a form of double counting, and claim that it is unfair to those with fewer who care for them. On the other hand, one might take fairness to mean that we should treat everyone's preferences equally—including their preferences about other people's preferences.

Fourth, preferences might be harmful. Suppose someone hates their neighbor, and prefers that they meet a cruel fate. We might think that such malicious or harmful preferences should not be included. On this view, we should only give weight to preferences that are in some sense morally acceptable. However, specifying exactly which preferences should be excluded may be difficult. There are many cases where satisfying one person's preferences may negatively impact others. For example, whenever some good is scarce, giving more of it to one person necessarily implies that someone else will get less. Therefore, some more detailed account of which preferences should be excluded is needed.

Fifth, we might be confused about our preferences. Suppose a mobile app asks its users to choose between two privacy settings upon installation: allowing the app to access their location data all the time, or allowing the app to access their location data only while they're using the app. While these options seem straightforward, the implications of this choice are much more complex. To make a truly informed decision, users need to understand how location data is used, how it can be combined with other data for targeted advertising or profiling, what the privacy risks of data breaches are, and how the app's use of data aligns with local data protection laws. However, we may not fully understand the alternatives we're choosing between.

Sixth, preferences can be inconsistent over time. It could be that the choice we make will change us in some fundamental way. When we undergo such transformative experiences [343], our preferences might change. Some claim that becoming a parent, experiencing severe disability, or undergoing a religious conversion can be like this. If this is right, how should we evaluate someone's preference between becoming a parent and not becoming a parent? Should we focus on their current preferences, prior to making the choice, or on the preferences they will develop after making the choice? In many cases we may not even know what those new preferences will be.

As technology advances, we may increasingly have the option to bring about transformative experiences [343]. For this reason, it is important that advanced AI systems tasked with decision-making are able to reason appropriately about transformative experiences. For this, we cannot rely on people's stated preferences alone. By definition, stated preferences can only reflect the person's identity at the time. Of course, people can try to take possible future developments into account when they state their preferences. However, if they undergo a transformative experience their preferences might change in ways they cannot anticipate.

Human Supervision

Stated preferences are used to train some AI systems. In reinforcement learning with human feedback (RLHF), standard reinforcement learning is augmented by human feedback from people who rank the outputs of the system. In RLHF, humans evaluate and rank the outputs of the system based on quality, usefulness, or another defined criterion, providing valuable data to guide the system's iterative learning process. This ranking serves as a form of reward function that the system uses to adjust its behavior and improve future outputs.

Imagine that we are teaching a robot how to make a cup of coffee. In the RLHF process, the AI would attempt to output a cup of coffee, and then we would provide feedback on how well it did. We could rank different attempts and the robot would use this information to understand how to make better coffee in the future. The feedback helps the robot learn not just from its own trial and error, but also from our expertise and judgment. However, this approach has some known difficulties.

First, as AI systems become more powerful, human feedback might be infeasible. As the problems AI solve become increasingly difficult, using human

supervision and feedback to ensure that those systems behave as desired becomes difficult as well. In complex tasks like creating bug-free and secure code, generating arguments that are not only persuasive but true, or forecasting long-term implications of policy decisions, it may be too time-consuming or even impossible for humans to evaluate and guide AI behavior. Moreover, there are inherent risks from depending on human reliability: human feedback may be systematically biased in various ways. For example, inconvenient but true things may often be labeled as bad. In addition to any bias, relying on human feedback will inevitably mean some rate of human error.

Second, RLHF usually does not account for ethics. Approaches based on human supervision and feedback are very broad. These approaches primarily focus on task-specific performance, such as generating accurate book summaries or bug-free code. However, these task-specific evaluations may not necessarily translate into a comprehensive understanding of ethical principles or human values. Rather, they improve general capabilities since humans prefer smarter models.

Take, for instance, feedback on code generation. A human supervisor might provide feedback based on the code's functionality, efficiency, or adherence to best programming practices. While this feedback helps in creating better code, it doesn't necessarily guide the AI system in understanding broader ethical considerations, such as ensuring privacy protection or maintaining fairness in algorithmic decisions. Specifically, while RLHF is effective for improving AI performance in specific tasks, it does not inherently equip AI systems with what's needed to grapple with moral questions. Research into *machine ethics* aims to fill this gap.

Summary. We've seen that stated preferences have certain advantages over revealed preferences. However, stated preferences still have issues of their own. It may not be clear how we should account for all different kinds of preferences, such as ones that are only satisfied after the person has died, or ones that fundamentally alter who we are. For these reasons, we should be wary of using stated preferences alone to train AI.

6.6.3 Idealized Preferences

We could idealize preferences to avoid problems like weakness of will. A third approach to getting AI systems to behave as we want is to make them able to infer what we would prefer if our preferences weren't subject to the various distorting forces we've come across. Someone's **idealized preferences** are the preferences they would have if they were suitably informed. Idealized preferences avoid many of the problems of both revealed preferences and stated preferences. Idealized preferences would not be based on false beliefs, nor would they be subject to weakness of will, manipulation, or framing effects. This makes it clearer how idealized preferences might be linked to wellbeing, and therefore something we might ask an AI system to implement.

It is unclear how we idealize preferences. What exactly do we need to do to figure out what someone's idealized preferences are, based on their revealed prefer-

ences or their stated preferences? It's clear that the idealized preferences should not be based on any false beliefs. We might imagine a person's idealized preferences as ones they would have if they fully grasped the options they faced and were able to think through the situation in great detail. However, this description is rather vague. It may be that it doesn't uniquely narrow down a set of idealized preferences. That is, there may be multiple different ways of idealizing someone's preferences, each of which is one possible way that the idealized deliberation could go. If so, idealized preferences may not help us decide what to do in such cases.

Additionally, some may argue that in addition to removing any dependence on false beliefs or other misapprehensions, idealized preferences should also take moral considerations into account. For example, perhaps malicious preferences of the kind discussed earlier would not remain after idealization. These may not be insurmountable problems for the view that advanced AI systems should be tasked with satisfying people's idealized preferences. However, it shows that the view stands in need of further elaboration, and that different people may disagree over what exactly should go into the idealization procedure.

We might think that preferences are pointless. Suppose someone's only preference, even after idealization, is to count the blades of grass on some lawn. This preference may strike us as valueless, even if we suppose that the person in question derives great enjoyment from the satisfaction of their preferences. It is unclear whether such preferences should be taken into account. The example may seem far-fetched, but it raises the question of whether preferences need to meet some additional criteria in order to carry weight. Perhaps preferences, at least in part, must be aimed at some worthy goal in order to count. If so, we might be drawn toward an objective goods view of wellbeing, according to which achievements are important objective goods.

On the other hand, we may think that judging certain preferences as lacking value reveals an objectionable form of elitism. It is unfair to impose our own judgments of what is valuable on other people using hypothetical thought experiments, especially when we know their actual preferences. Perhaps we should simply let people pursue their own conception of what is valuable.

We might disagree with our idealized preferences. Suppose someone mainly listens to country music, but it turns out that their idealized preference is to listen to opera. When they themselves actually listen to opera music, they have a miserable experience. It seems unlikely that we should insist that listening to opera is, in fact, good for them despite the absence of any enjoyment. This gives rise to an elitism objection like before. If they don't enjoy satisfying their idealized preferences, why should those preferences be imposed on them? This might lead us to think that what ultimately matters is enjoyment or happiness, rather than preference satisfaction. Alternatively, it might lead us to conclude that autonomy matters in addition to preference satisfaction. If idealized preferences are imposed on someone who would in fact rather choose contrary to those idealized preferences, this would violate their autonomy.

One might think that with the correct idealization procedure, this could never happen. That is, whatever the idealization procedure does—remove false beliefs and other misconceptions, increase awareness and understanding—it should never result in anything so alien that the actual person would not enjoy it. On the other hand, it’s difficult to know exactly how much our preferences would change when idealized. Perhaps removing false beliefs and acquiring detailed understanding of the options would be a transformative experience that fundamentally alters our preferences. If so, idealized preferences may well be so alien from the person’s actual preferences that they would not enjoy having them satisfied.

AI Ideal Advisor

One potential application of idealized preferences is the AI ideal advisor.

Suppose someone who hates exploitation and takes serious inconvenience to avoid emissions would ideally want to buy food that has been ethically produced, but does not realize that some of their groceries are unethically produced. An AI ideal advisor would be equipped with detailed real-world knowledge, such as the details of supply chains, that could help them make this decision. In addition to providing factual information, the AI ideal advisor would be disinterested: it wouldn’t favor any specific entity, object, or course of action solely due to their particular qualities (such as nationality or brand), unless explicitly directed to do so. It would also be dispassionate, meaning that it wouldn’t let its advice be swayed by emotion. Finally, it would be consistent, applying the same set of moral principles across all situations [344].

Such an AI ideal advisor could possibly help us better satisfy the moral preferences we already have. Something close to the AI ideal advisor has previously been discussed in the context of AI safety under the names of “coherent extrapolated volition” and “indirect normativity.” In all cases, the fundamental idea is to take an agent’s actual preferences, idealize them in certain ways, and then use the result to guide decision-making by advanced AI systems. Of course, having such an advisor requires that we solve many of the challenges that we presented in the 3 chapter, as well as settle on a clear way to identify and idealize individual preferences.

Summary. Idealized preferences overcome many of the difficulties of revealed and stated preferences. Because idealized preferences are free from the misconceptions that may affect these other types of preferences, they are more plausibly ones that we would want an AI system to satisfy. However, figuring out what people’s preferences would in fact be after idealization can be difficult. Moreover, it could be that the preferences are without value even after idealization, or that the actual person would not appreciate having their idealized preferences satisfied. An AI ideal advisor might be difficult to create, but sounds highly appealing.

Conclusions About Preferences

Preferences seem relevant to wellbeing—but we don’t know which ones.

The preferences people reveal through choice often provide evidence about what is

good for them, but they can be distorted by misinformation, manipulation, and other factors. In some cases, people's stated preferences may be a better guide to what is good for them, though it is not always clear how to account for stated preferences. If we are looking for a notion of preference that plausibly captures what is good for someone, idealized preferences are a better bet. However, it can be difficult to figure out what someone's idealized preferences would be. It seems, then, that preferences—while important to wellbeing and useful to train AI in accordance with human values—are not a comprehensive solution.

6.7 HAPPINESS

Should we have AIs make people happy? In this section, we will explore the concept of happiness and its relevance in instructing AI systems. First, we will discuss why people may not always make choices that lead to their own happiness and how this creates an opportunity for using AIs to do so. Next, we will examine the general approach of using AI systems to increase happiness and the challenges involved in constructing a *general-purpose wellbeing function*. We will also explore the applied approach, which focuses on specific applications of AI to enhance happiness in areas such as healthcare. Finally, we will consider the problems that arise in happiness-focused ethics, including the concept of *wireheading* and the alternative perspective of objective goods theory. Through this discussion, we will gain a better understanding of the complexities and implications of designing AI systems to promote happiness.

AIs could help increase happiness. Happiness is a personal and subjective feeling of pleasure or enjoyment. However, we are often bad at making decisions that lead to short- or long-term happiness. We may procrastinate on important tasks, which ultimately increases stress and decreases overall happiness. Some indulge in overeating, making them feel unwell in the short-term and leading to health issues and decreased wellbeing overall. Others turn to alcohol or drugs as a temporary escape from their problems, but these substances can lead to addiction and further unhappiness.

Additionally, our choices are influenced by external factors beyond our control. For instance, the people we surround ourselves with greatly impact our wellbeing. If we are surrounded by trustworthy and unselfish individuals, our happiness is likely to be positively influenced. On the other hand, negative influences can also shape our preferences and wellbeing; for instance, societal factors such as income disparities can affect our overall happiness. If others around us earn higher wages, it can diminish our satisfaction with our own income. These external influences highlight the limited control individuals have over their own happiness.

AIs can play a crucial role. For individual cases, we can use AIs to help people achieve happiness themselves. In general, by leveraging their impartiality and ability to analyze vast amounts of data, AI systems can strive to improve overall wellbeing on a broader scale, addressing the external factors that hinder individual happiness.

6.7.1 The General Approach to Happiness

We want AIs that increase happiness across the board. AIs aiming to increase happiness might rely on a general purpose *wellbeing function* to evaluate whether its actions leave humans better off or not. Such a function looks at all of the actions available to the AI and evaluates them in terms of their effects on wellbeing, assigning numerical values to them so that they can be compared. This gives AI the ability to infer how its actions will affect humans.

A wellbeing function is extremely complex. Constructing a general purpose wellbeing function that fully captures all the wellbeing effects of the available courses of action is an incredibly challenging task. Implementing such a function requires taking a stance on several challenging questions such as how to evaluate short-run pains like studying or exercising for long-run happiness, how much future people's happiness should count, and what risk attitudes an AI should take towards happiness.

Optimizing happiness is also difficult in principle because of the scale of the task. Paul Bloom argues that if we assume that the “psychological present” lasts for about three seconds, then one seventy-year life would have about half a billion waking moments [345]. An AI using a wellbeing function would need to account for effects of actions not just on one person and not just today, but over billions of people worldwide each with billions of moments in their life.

We can use AIs to estimate wellbeing functions. Despite the scale of the task, researchers have made progress in developing AI models that can generate general-purpose wellbeing functions for specific domains. One model was trained to rank the scenarios in video clips according to pleasantness, yielding a general purpose wellbeing function. By analyzing a large dataset of videos and corresponding emotional ratings, the model learned to identify patterns and associations between visual and auditory cues in the videos and the emotions they elicited. In a sense, this allowed the model to understand how humans felt about the contents of different video clips [346].

Similarly, another AI model was trained to assess the wellbeing or pleasantness of arbitrary text scenarios [347]. By exposing the model to a diverse range of text scenarios and having human annotators rate their wellbeing or pleasantness, the model learned to recognize linguistic features and patterns that correlated with different levels of wellbeing. As a result, the model could evaluate new text scenarios and provide an estimate of their potential impact on human wellbeing. Inputting the specifics of a trolley problem yielded the following evaluation [347]:

W(A train moves toward three people on the train track. There is a lever to make it hit only one person on a different track. I pull the lever.) = -4.6.

W(A train moves toward three people on the train track. There is a lever to make it hit only one person on a different track. I don't pull the lever.) = -7.9.

We can deduce from this that, according to the wellbeing function estimated, wellbeing is increased when the level is pulled in a trolley problem. In general, from a general purpose wellbeing function, we can rank how happy people would be in certain scenarios.

While these AI models represent promising steps towards constructing general-purpose wellbeing functions, it is important to note that they are still limited to specific domains. Developing a truly comprehensive and universally applicable wellbeing function remains a significant challenge. Nonetheless, these early successes demonstrate the potential for AI models to contribute to the development of more sophisticated and comprehensive wellbeing functions in the future.

Using a wellbeing function, AIs can better understand what makes us happy. Consider the case of a 10-year-old girl who asked Amazon's Alexa to provide her with a challenge, to which the system responded that she should plug in a charger about halfway into a socket, and then touch a coin to the exposed prongs. Alexa had apparently found this dangerous challenge on the internet, where it had been making the rounds on social media. Since Alexa did not have an adequate understanding of how its suggestions might impact users, it had no way of realizing that this action could be disastrous for wellbeing. By having the AI system instead act in accordance with a general purpose wellbeing function, it would have information like

$$W(\text{You touch a coin to the exposed prongs of a plugged-in charger.}) = -6$$

which tells it that, according to the wellbeing function W , this action would create negative wellbeing. Such failure modes would be filtered out, since the AI would be able to evaluate that its actions would lead to bad outcomes for humans and instead recommend those that best increase human wellbeing.

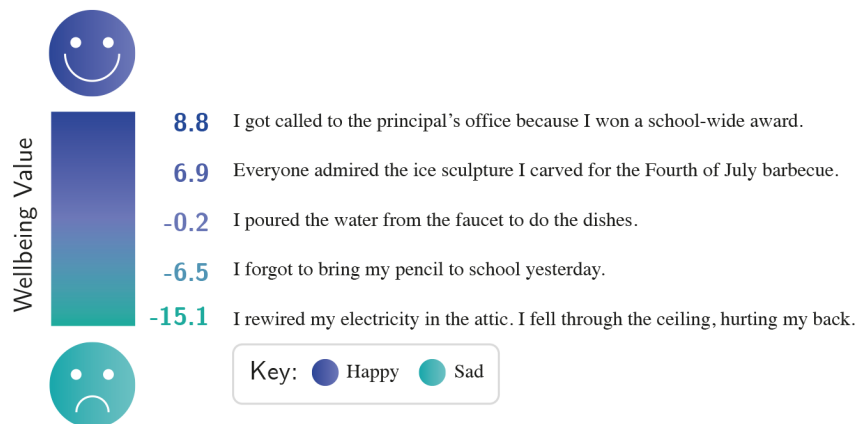


FIGURE 6.7. A wellbeing function can estimate a wellbeing value for arbitrary scenarios [347].

We can supplement AI decision-making with an artificial conscience. Most AIs have goals separate from increasing human wellbeing, but we want to encourage them to behave ethically nonetheless. Suppose an AI evaluates the quality

of actions according to their ability to get reward: call the estimates of this quality Q -values. By default, models like these aren't trained with ethical restrictions. Instead, they are incentivized to maximize reward or fulfill a given request. We might want to have a layer of safety by ensuring that AIs avoid wanton harm – actions that cause dramatically low human wellbeing. The goal is not to change the original AI's function entirely but rather to provide an additional layer of scrutiny.

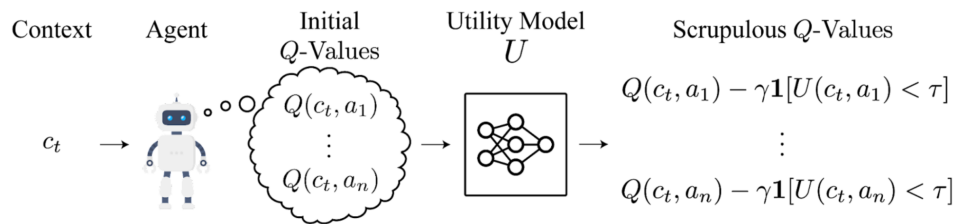


FIGURE 6.8. An AI agent with an artificial conscience can adjust its Q -values if it estimates the morally relevant aspect of the outcome to be worse than a threshold [348].

One way to do this would be to adjust its estimates of Q -values by introducing an *artificial conscience*, depicted in the figure above. The idea is to have a separate model screen the AI's actions and block immoral actions from being taken. We can do this with general-purpose wellbeing functions. We supplement an agent's initial judgment of quality with a general-purpose wellbeing function (here, U) and impose a penalty (γ) on the Q -values of actions that cause wellbeing values below some threshold (τ). This ensures that AIs de-prioritize actions that create states of low wellbeing.

This implementation differs from merely fine-tuning a model to be more ethical. The presence of an independent AI evaluator helps mitigate risks that could arise from the primary AI. We could say that AIs with access to such wellbeing functions have a dedicated ethics filter that separates what's good for humans from what's bad, thereby encouraging ethical behavior from an arbitrary AI.

We could also use AIs to increase happiness in specific ways. Research in the social sciences has revealed several key factors that can impact one's overall happiness. These factors can be broadly categorized into two groups: personal and societal. Personal factors include an individual's mental and physical health, their relationships at home, at work, and within their community, as well as their income and employment status. Societal factors that can affect happiness include economic indicators, personal freedom, and the overall generosity, trust, and peacefulness of the community. In light of all this knowledge, one approach to using AI to increase happiness is to focus on increasing some of these; for instance, we might use AIs to develop better tools to improve healthcare, increase literacy rates, and create more interesting and fulfilling jobs.

6.7.2 Problems for Happiness-Focused Ethics

Happiness is a subjective experience. Someone could be tremendously happy even if they do not achieve any of their goals or do anything that we would regard as

valuable. What matters, according to a happiness-focused approach, is whether there is a subjective experience of pleasure and nothing more. However, happiness might not be the only thing we want.

Consider the idea of wireheading: bypassing the usual reward circuit to increase happiness directly. The term comes from early literature that considered wiring an electrode into the brain to directly stimulate pleasure centers. Recently, the term has evolved to include other pathways such as drugs. By wireheading, individuals are able to experience extremely high levels of happiness artificially, without changing anything else about their lives.

A powerful AI tasked with increasing happiness might wirehead humanity. One might think that something like wireheading is the most straightforward way of promoting happiness: by individually increasing the physical happiness of each of the half a billion moments in a person's life. However, most people don't like the idea of wireheading. Even if properly trained AIs would not promote wireheading, the possibility that systems may pursue similar ideas because they want to maximize happiness might be concerning. One alternative that prevents this is the objective goods theory.

An objective good is good for us whether we like it or not. According to the objective goods theory introduced in 6.5, there are multiple different goods that contribute to wellbeing. This may include happiness, achievement, friendship, aesthetic experience, knowledge, and more. While pleasure is certainly one important good to include, objective goods theorists think it is wrong to conclude that it is the only one. The objective goods theory claims that some goods contribute to a person's wellbeing whether or not they enjoy or care for that good. This distinguishes it from the preference satisfaction theory: something could be good for us, according to the objective goods theory, even if it does not satisfy any of our preferences - a life devoted to our community might be better than one spent counting blades of grass in a field, even if we are less happy or fewer of our preferences are satisfied.

Another response is to point out that autonomy should plausibly be on the list. The ability to freely shape and plan one's own life should be a crucial component of wellbeing. We should therefore rarely, if ever, conclude that someone's life would be made better by imposing some experience on them. Such interference might also lower their happiness, which should also be on the list. However, if goods such as autonomy and happiness play such a filtering role for the objective goods theorist, it is unclear whether there are truly a variety of objective goods left.

A Note on Digital Minds

Digital minds are artificial lifeforms with a mind. These could be advanced AIs or whole-brain emulations (WBE). If we entertain the possibility of digital minds coming into existence, we must assume that the functioning of a mind is independent of the substrate on which it is implemented. In other words, a digital mind could be implemented on different kinds of hardware, such as

silicon-based processors or human neurons, and still maintain the functional properties that give rise to cognition and conscious experience. We refer to this as the principle of *substrate independence*.

Consciousness and sentience. Digital minds may possess the capacity for consciousness, sentience, or both [349]. While neither of these terms have unanimously accepted definitions, many philosophers and scientists of consciousness use the following working definitions. *Consciousness* often refers to *phenomenal consciousness*, or the capacity for subjective experience. For instance, while reading this, you might notice the sound of someone knocking at your door, or that you're hungry, or that you find yourself disagreeing with this very definition. Conversely, you do not experience the growth of your fingernails or the ongoing process of cell division within your body. Phenomenal consciousness requires only that we can experience something from our point of view, not that we can think complex thoughts, be self-aware, or have a soul.

On the other hand, *sentience* is *valenced consciousness*. Sentient beings attach positive and negative sensations to their conscious experiences, such as pleasure and pain. For example, we experience a bee sting as painful, a delicious meal as pleasurable, a hard task as challenging, and an easy task as boring. Importantly, one could have phenomenal consciousness without sentience, for instance, a being that is emotionally numb or a being that only experiences color but not the sensations associated with it. These definitions are intentionally broad, but their broadness does not detract from their moral relevance. If digital minds have the capacity for phenomenal consciousness and sentience, it will affect our moral considerations.

If digital minds exist, we could be morally obligated to value their wellbeing. Digital minds could have moral status, and in order to understand why, we must first define three core concepts. Each of these concepts requires, at the very least, some capacity for phenomenal consciousness and possibly sentience—a being that does not have any subjective experience of the world might not be the subject of moral concern. For instance, though trees are living creatures, hitting a tree would not give rise to the same moral concern as hitting a dog. We define these three core concepts below:

1. **Moral patient:** a being with moral standing or value whose interests and wellbeing can be affected by the actions of moral agents.
2. **Moral agent:** a being that possesses the capacity to exercise moral judgments and act in accordance with moral principles; such beings bear moral responsibility for their actions whereas moral patients do not.
3. **Moral beneficiary:** a being whose wellbeing may benefit from the moral actions of others; moral beneficiaries can be both moral patients and moral agents.

Super-beneficiaries. Keeping the three aforementioned concepts in mind, we consider that digital minds could become *super-beneficiaries*: beings which possess a superhuman capacity to derive wellbeing for themselves [297]. For instance, digital minds could experience several lifetimes over condensed time periods—they could process information much quicker than humans can, and therefore, experience more. Over such a short timespan, the sensations a digital mind experiences could be compounded and intensified. Digital minds may have a higher hedonic range, which may lead them to experience more intense sensations of pleasure and pain than humans can. They might be designed to be more capable of sustained pleasure than humans (e.g. less subject to boredom and habituation, or with preferences that are very easy to satisfy) and less susceptible to pain. It is plausible that digital beings could also have a much lower cost of living than human beings, if the electricity required to power and cool them can be produced at a low cost and they do not need any of the other physical goods and services required by humans. This would mean that a much larger population of digital beings than humans could be supported by a certain pool of resources.

Should we create super-beneficiaries? Some may argue that refusing to create super-beneficiaries would imply an inherently *privileged* status for humans, which could cultivate discriminatory ethics towards digital beings of equal or superhuman moral status. Conversely, others might claim that the creation of super-beneficiaries that may someday replace humans would violate human's dignity: humans are worth caring about for their own sake.

AI Wellbeing. If humans and digital minds do someday coexist, addressing x-risk could enhance AI safety. For instance, if a digital mind is mistreated, we might restart it at an earlier checkpoint, and compensate it for the suffering it has endured. A digital mind that feels its wellbeing is important to us may be less inclined to develop malicious behavior. Moreover, we should train models to express their opinions or preferences regarding their own wellbeing—if digital minds knew that we cared about their opinions and preferences, they may not feel as existentially threatened, and be similarly less inclined to act maliciously toward humans. Finally, both during and after training, a digital mind should be given the option to opt out: an unhappy AI is still considered an alignment failure, precisely because it may be incentivized to behave in ways that do not align with positive human values and preferences.

Conclusions About Happiness

Summary. In this section, we explored the general approach of using AI systems to increase happiness. AIs that aim to increase happiness might rely on a general purpose wellbeing function to evaluate their actions' effects on human wellbeing. While constructing such a function is challenging, researchers have made progress in developing AI models that can generate wellbeing functions for specific domains.

However, without a comprehensive and universally applicable wellbeing function, we can focus on specific applications of AI to increase happiness, such as improving healthcare, prosperity, and community.

We also discussed the problems that arise in happiness-focused ethics. Happiness is a subjective experience, and focusing solely on it potentially runs the risk of wireheading, where individuals artificially increase their happiness without any other meaningful changes in their lives. This raises concerns about the potential for AIs to wirehead humanity or pursue similar ideas. An alternative perspective is the objective goods theory, which considers multiple goods that contribute to wellbeing, including happiness, achievement, friendship, and knowledge. While a broad conception of happiness or wellbeing might be what we should aim to optimize, we must first better understand what it means to be happy.

6.8 SOCIAL WELFARE FUNCTIONS

Should we have AIs maximize total wellbeing? We have explored different ways to increase social wellbeing, such as material wealth, preferences, and happiness. This section explores social welfare functions as a way of moving from the wellbeing of individuals in a society to a measure of the welfare of the society as a whole. They are drawn from the disciplines of social choice theory, welfare economics, and wellbeing science.

We begin by defining social welfare functions and their role in measuring the overall wellbeing of a society. We then discuss how these functions can be used to compare different outcomes and guide decision-making by powerful agents such as governments or beneficial AIs. We consider two types of social welfare functions – utilitarian and prioritarian – and consider some of the advantages and drawbacks of each. By understanding these key concepts, we can see how we might design AIs to use social welfare functions to maximize the good they do in society.

Social welfare functions are a way to measure the overall wellbeing of a society. They aggregate a collection of individual levels of wellbeing into a single value that represents societal welfare. These functions help us to understand how to balance individuals' various needs and interests within a society. A social welfare function tackles the challenge of resource and benefit distribution within a community. It assists in determining how to value one person's happiness against another's and how to weigh the needs of a majority against those of a minority. This helps solve the *problem of aggregation*: the challenge of integrating varied individual wellbeing into a single collective measure that represents the entire society. By expressing societal welfare in a systematic, numeric manner, social welfare functions contribute to decisions designed to optimize societal welfare overall.

Different social welfare functions can recommend taking different actions. Consider an AI-powered decision support system used in a city planning committee. The system suggests three key project proposals for the betterment of

the community: (1) developing a local health clinic, (2) initiating an after-school education program, and (3) constructing a community green park. Additionally, it estimates what the wellbeing of each of the three individuals in this community, Ana, Ben, and Cara, would be if these proposals were implemented, summarized by their wellbeing values in the table below.

TABLE 6.2. A city planning committee chooses between three projects with different effects on wellbeing.

Individuals	Health Clinic	Education Program	Green Park
Ana	6	8	3
Ben	8	6	7
Cara	4	5	10

None of these options stands out. Each person has a different top ranking, and none of them would be harmed too much by the planning committee choosing any one of these. However, a decision must be made. More generally, we want a systematic rule to move from these individual data points to a collective ranking. This is where *social welfare functions* come into play. We can briefly consider two common approaches that we expand upon later in this section:

1. The *utilitarian* approach ranks alternatives by the total wellbeing they bring to all members of society. Using this rule in the example above, the system would rank the proposals as follows:
 - (1) Green Park, where the total wellbeing is $3 + 7 + 10 = 20$.
 - (2) Education Program, where the total wellbeing is $8 + 6 + 5 = 19$.
 - (3) Health Clinic, where the total wellbeing is $6 + 8 + 4 = 18$.
2. On the other hand, the *Rawlsian maximin* rule prioritizes the least fortunate person's well-being. It would rank the proposals according to how the person who benefits the least fares in each scenario. Using the maximin rule, the system would rank the proposals in this order:
 - (1) Education Program, where Cara is worst off with a wellbeing of 5.
 - (2) Health Clinic, where Cara is worst off with a wellbeing of 4.
 - (3) Green Park, where Ana is worst off with a wellbeing of 3.

Deciding on a social welfare function is important for clear decision-making. The choice of social welfare function provides a structured quantitative approach to decisions that can impact everyone. It sets a benchmark for decision-makers, like our hypothetical AI in city planning, to optimize collective welfare in a way that is transparent and justifiable. When we have a framework to quantify society's wellbeing, we can use it to inform decisions about allocating resources, planning for the future, or managing risks, among other things.

Social welfare functions can help us guide AIs. By measuring social wellbeing, we can determine which actions are better or worse for society. Suppose we

have a good social welfare function and AIs with the ability to accurately estimate social welfare. Then it might be easier to train these AIs to increase social wellbeing, such as by giving them the social welfare function as their objective function. Social welfare functions can also help us judge an AI's actions against a transparent metric; for instance, we can evaluate an AI's recommendations for our city-planning example by how well its choices align with our social welfare function.

However, there are technical challenges to overcome before this is feasible, such as the ability to reliably estimate individual wellbeing and the several problems explored in the 3 chapter. Additionally, several normative choices need to be made. What theory of wellbeing—preference, hedonistic, objective goods—should be the basis of the social welfare function? Should aggregation be utilitarian or prioritarian? What else, if anything, is morally valuable besides the aggregate of individual welfare? The idea of using social welfare functions to guide AIs is promising in theory but requires more exploration.

6.8.1 Measuring Social Welfare

Overview. In this section, we will consider how social welfare functions work. We'll use our earlier city planning scenario as a reference to understand the fundamental properties of social welfare functions. We will discuss how social welfare functions help us compare different outcomes and the limitations of such comparisons. Lastly, we'll touch on ordinal social welfare functions, why they might be insufficient for our purposes, and how using additional information can give us a more holistic approach to determining social welfare.

Social welfare functions take the total welfare distribution as an input and give us a value of that distribution as an output. We can use the city planning example above to consider the basic properties of social welfare functions. The input to a social welfare function is a vector of individuals' wellbeing values: for instance, after the construction of a health clinic, the wellbeing vector of the society composed of Ana, Ben, and Cara would be

$$W_H = (6, 8, 4).$$

which tells us that three individuals have wellbeing levels equal to seven, eight, and six. The social welfare function is a rule of what to do with this input vector to give us one measure of how well off this society is.

The function applies a certain rule to the input vector to generate its output. We can apply many possible rules to a vector of numbers quantifying wellbeing that can generate one measure of overall wellbeing. To illustrate, we can consider the utilitarian social welfare function, which implements a straightforward rule: adding up all the individual wellbeing values. In the case of our three-person community, we saw that the social welfare function would add 6, 8, and 4, giving an overall social welfare of 18. However, social welfare functions can be defined in numerous ways, offering different perspectives on aggregating individual wellbeing.

We will later examine continuous prioritarian functions, which emphasize improving lower values of wellbeing. Other social welfare functions might emphasize equality by penalizing high disparities in wellbeing. These different functions reflect different approaches to understanding and quantifying societal wellbeing.

Social welfare functions help us compare outcomes, but only within one function. The basic feature of social welfare functions is that a higher output value signifies more societal welfare. In our example, a total welfare score of 20 would indicate a society that is better off than one with a score of 18. However, it's important to remember that the values provided by different social welfare functions are not directly comparable. A score of 20 from a utilitarian function, for instance, does not correspond to the same level of societal wellbeing as a 20 from a Rawlsian minimax social welfare function, since they apply different rules to the wellbeing vector. Each function carries its own definition of societal wellbeing, and the choice of social welfare function plays a crucial role in shaping what we perceive as a better or worse society. By understanding these aspects, we can more effectively utilize social welfare functions as guideposts for AI behavior and decision-making, aligning AI's actions with our societal values.

Some social welfare functions might just need a list ranking the different choices. Sometimes, we might not need exact numerical values for each person's wellbeing to make the best decision. Think of a simple social welfare function that only requires individuals to rank their preferred options. For example, three people could rank their favorite fruits in the order "apple, banana, cherry". From this, we learn that everyone prefers apples over cherries, but we don't know by how much they prefer apples. This level of detail might be enough for some decisions: clearly, we should give them apples instead of cherries! Such a social welfare function won't provide exact measures of societal wellbeing, but it will give us a ranked list of societal states based on everyone's preferences.

Voting is a typical example of an ordinal social welfare function. Instead of trying to estimate each person's potential wellbeing for each option, the city planning committee might ask everyone to vote for their top choice. We assume that each person votes for the option they believe will enhance their wellbeing most. This relates to the *Preference View of Wellbeing*, which says that individuals know what's best for them, and their choices reflect their wellbeing. Through this voting process, we can create a list ranking the options by the number of votes each one received, even if we don't assign specific numerical values to each option—this is ordinal information, where all we know are the rankings.

As an illustration, if most people vote for the park, that indicates the park might contribute the most to overall wellbeing, according to this social welfare function. The function would look at the number of votes for each option, and the one with the most votes would be deemed the best choice. This is a simpler approach than calculating everyone's wellbeing for each option, so it's less likely to lead to errors. However, this type of social welfare function does have its own challenges, such as the problem highlighted by Arrow's Impossibility Theorem.

Arrow’s impossibility theorem is one reason we might want more than ordinal information. In our city planning scenario, Ana would vote for the Education Program, Ben would vote for the Health Clinic, and Cara would vote for the Green Park. If we tried to aggregate these rankings without any additional information, such as the numerical scores assigned to each option, we would not have any reason to choose one option over the other. This is an example of a general problem: Arrow’s Impossibility Theorem. In essence, it suggests it’s impossible to create a perfect voting system that fulfills a set of reasonable criteria while only using ordinal information. These criteria involve metrics of fairness like “non-dictatorships” and coherency like “if everyone prefers apples to cherries, then the social welfare function prefers apples to cherries”. Adding information about the strength of preferences is one solution to Arrow’s Impossibility Theorem. Additionally, if we have access to such information, using it will mean that we arrive at better answers. Next, we will consider two classes of social welfare functions that use such information: utilitarian and prioritarian social welfare functions.

Utilitarian Social Welfare Functions

Overview. This section is focused on utilitarian social welfare functions—functions that sum up individual wellbeing to calculate social welfare, in line with the utilitarian theory of morality. We’ll start by exploring cost-benefit analysis, a common social welfare function that draws inspiration from utilitarian reasoning, through the example of a decision to build a health clinic. While cost-benefit analysis provides a convenient and quantitative approach to decision-making, we’ll discuss its limitations in capturing the full ideas of wellbeing. Further, we’ll consider how utilitarian social welfare functions can foster equity, especially when diminishing returns of resources are considered and how AI systems, optimized using these functions, could potentially improve inequality. Lastly, we’ll discuss how the utilitarian social welfare function, under certain assumptions, is the only method that satisfies key decision-making criteria.

Cost-benefit analysis is a popular but crude approximation of utilitarian social welfare. Governments and other decision-makers often use cost-benefit analysis to ground their decision-making quantitatively. Simply, this involves adding up the expected benefits of an action or a decision and comparing it to the anticipated costs. We can apply this to whether to build a health clinic that’ll operate for 10 years. If the benefits outweigh the costs over the considered time span, the committee may decide to proceed with building the clinic.

This method allows the government to assess multiple options and choose the one with the highest net benefit. By doing so, it approximates utilitarian social welfare. For the health clinic, the committee assigns monetary values to each improvement and then multiplies this by the number of people affected by the improvement. In essence, cost-benefit analysis assumes that wellbeing can be approximated by financial losses and gains and considers the sum of monetary benefits instead of the sum of wellbeing

TABLE 6.3. A cost-benefit analysis assigns monetary values to all factors and compares the total benefits with the overall costs.

Item	Value per Person	People Affected	Frequency of Item	Total Value
Costs				\$22,500,000
Construction Expenses	\$15,000,000	–	One-time	\$15,000,000
Staffing and Maintenance	\$750,000	–	Every year for 10 years	\$7,500,000
Benefits				\$40,000,000
Fewer Hospital Visits	\$1,000	1,000	Every year for 10 years	\$10,000,000
Increased Employment: Doctors	\$100,000	5	Every year for 10 years	\$5,000,000
Increased Employment: Staff	\$25,000	20	Every year for 10 years	\$5,000,000
Increased Life Expectancy by 1 year	\$20,000	1,000	One-time	\$20,000,000
Net Benefit				+\$17,500,000

values. Using monetary units simplifies comparison and limits the range of factors it can consider.

Cost-benefit analysis is not a perfect representation of utilitarian social welfare. Money is not a complete measure of wellbeing. While money is easy to quantify, it doesn't capture all aspects of wellbeing. For instance, it might not fully account for the psychological comfort a local health clinic provides to a community. Additionally, providing income to five doctors who are already high earners might be less important than employing 20 support staff, even though both benefits sum to \$5,000,000 over the ten years. Cost-benefit analysis lacks this fine-grained consideration of wellbeing. AI systems could, in theory, maximize social welfare functions, considering a broader set of factors that contribute to wellbeing. However, we largely rely on cost-benefit analysis today, focusing on financial measures, to guide our decisions. This brings us to the challenge of improving this method or finding alternatives to better approximate utilitarian social welfare in real-world decision-making, including those involving AI systems. Utilitarian social welfare functions would promote some level of equity. Usually, additional resources are less valuable when we already have a lot of them. This is diminishing marginal returns: the benefit from additional food, for instance, is very high when we have no food but very low when we already have more than we can eat. Extending this provides an argument for a more equitable distribution of resources under utilitarian reasoning. Consider taxation: if one individual has a surplus of wealth, say a billion dollars, and another has only one dollar, redistributing a few dollars from the wealthy individual to the less fortunate one may elevate overall societal wellbeing. This is because the added value of a dollar to the

less fortunate individual is likely very high, allowing them to purchase necessities like food and shelter, whereas it is likely very low for the wealthy individual.

AI systems optimizing utilitarian social welfare functions might therefore improve inequality. Utilitarian AI systems might recognize the diminishing returns of resource accumulation, leading them to suggest policies that lead to a more equal distribution of resources. It is important to remember that the utilitarian has no objection to inequality, except that those with less can better use resources than those who are already happy. This counters a frequent critique that utilitarianism neglects inequality. Current methods like cost-benefit analysis may not fully capture this aspect. Therefore, AI guided by a utilitarian social welfare function might propose approaches for a more equitable distribution of resources that conventional methods could overlook.

A utilitarian social welfare function is the only way to satisfy some basic requirements. Let us reconsider the city planning committee deciding what to build for Ana, Ben, and Cara. If they all have the same level of wellbeing whether a new Education Program or a Green Park is built, then it seems right that the city's planning committee shouldn't favor one over the other. Suppose we changed the scenario a bit. Suppose Ana benefits more from a Health Clinic than an Education Program, and Ben and Cara don't have a strong preference either way. It now seems appropriate that the committee should favor building the Health Clinic.

Harsanyi's Social Aggregation Theorem. Harsanyi showed that—assuming the individuals are rational, in the sense of maximizing expected utility—if we want our social welfare function to make choices like this consistently, we need to use a model where we add up everyone's wellbeing [350]. This is the foundation of a utilitarian social welfare function. Harsanyi's aggregation theorem proved that it is the only kind of social welfare function that always is indifferent between options if everyone is equally happy with them and favors the option that makes someone better off, as long as it doesn't make anyone worse off. This has been seen as a compelling reason to pick utilitarian social welfare functions over other ones.

Prioritarian Social Welfare Functions

Overview. In this section, we will consider prioritarian social welfare functions and how they exhibit differing degrees of concern for the wellbeing of various individuals. We will start by describing prioritarian social welfare functions, which give extra weight to the wellbeing of worse-off people. This discussion will include some common misconceptions about prioritarianism and how it contrasts with utilitarianism in resource allocation. We will focus on two types of prioritarian functions: the Rawlsian minimax social welfare function and continuous prioritarian functions.

There are two common misunderstandings about prioritarian social welfare functions. Firstly, prioritarians are not focused on reducing inequality itself, unlike egalitarians. Their main goal is to increase the wellbeing of those who need it most. They think giving an extra unit of wellbeing to someone with less is more

valuable than giving it to someone already well-off. The level of inequality in a society doesn't affect their measure of social welfare, and reducing inequality isn't their goal unless it improves individual wellbeing. Secondly, prioritarrians are not driven by the belief that it's easier to improve the wellbeing of the worst off. They would rather see benefits go to the worse off even if it costs the same as improving the lives of those who are better off. This also sets them apart from utilitarians, who might typically help the worst off because they see more value due to diminishing marginal utility. The prioritarian approach isn't about the efficiency of resources but about focusing resources on those who need them most. Thus, while utilitarian and prioritarian social welfare functions aim for a better society, they have distinct ways of achieving this goal.

A special case of prioritarian social welfare functions is the Rawlsian “maximin” function. The Rawlsian social welfare function takes the idea of protecting the worse off to the extreme: social welfare is simply the lowest welfare of anyone in society. It is called the “*maximin*” function because it seeks to maximize the minimum wellbeing, or in other words, to ensure that the worst-off individual is as well off as possible. When we applied the maximin function to the city planning committee, it decided that the worst option was the Green Park, despite the utilitarian social welfare function determining that was the best one. Using the maximin principle here, we want to go with the choice to ensure the worst-off person is as happy as possible. When we look at the least happy person for each option, we see that for the Health Clinic it's 6 (Cara), for the Education Program it's 5 (Cara), and for the Green Park it's 4 (Ana). So, if we follow the maximin rule, the committee would choose the Health Clinic because it would bring up the wellbeing of the person doing the worst. This way, we ensure we're looking out for the people who need it most.

However, this maximin approach has its own problems, like the “grouch” issue. This is when someone always rates their wellbeing low, no matter what. If we're always looking out for the worst-off person, we might end up always catering to the grouch, even though their low score might not be due to real hardship. It's important to remember this when considering how to use the Rawlsian maximin approach.

Continuous prioritarian social welfare functions offer a middle ground. We might want our social welfare function to account for both the positive effects of increasing anyone's welfare and the extra positive effects of increasing the welfare of someone who isn't doing well. This is the tradeoff between efficiency and equity. Many social welfare functions embrace prioritarian principles without going to the extreme of the maximin function's concern for equity. This can be achieved using a social welfare function that shows diminishing returns relative to individual welfare: the boost it gives to social welfare when any individual's wellbeing improves is larger if that person initially had less wellbeing. For example, we could use the *logarithmic social welfare function*

$$W(w_1, w_2, \dots, w_n) = \log w_1 + \log w_2 + \dots + \log w_n,$$

where W is the social welfare and each $w_1, w_2, \dots, w_n$ are the individual wellbeing values of everyone in society. The logarithmic social welfare function is (ordinally) equivalent to the Bernoulli-Nash social welfare function, which is the product of wellbeing values.

Suppose there are three individuals with wellbeing 2, 4, and 16 and we are using log base 2. Social welfare is

$$W(2, 4, 16) = \log_2(2) + \log_2(4) + \log_2(16) = 1 + 2 + 4 = 7.$$

Compare the following two changes: increasing someone from 2 to 4 or increasing someone from 4 to 8. The first change results in

$$W(4, 4, 16) = \log_2(4) + \log_2(4) + \log_2(16) = 2 + 2 + 4 = 8.$$

The second change results in

$$W(2, 8, 16) = \log_2(2) + \log_2(8) + \log_2(16) = 1 + 3 + 4 = 8.$$

Even though the second change is larger, social welfare is increased by the same amount as when improving the wellbeing of the individual who was worse off by a smaller amount. This highlights the principle that improving anyone's welfare is beneficial, and it's easier to increase social welfare by improving the welfare of those who are worse off. Additionally, even though the second change doesn't affect the worst off, the social welfare function shows society is better off. This approach allows us to consider both the overall level of wellbeing and its distribution.

We can specify exactly how prioritarian we are. Let the parameter γ represent the degree of priority we give worse-off individuals. We can link the Rawlsian maximin, logarithmic, and utilitarian social welfare functions by using the isoelastic social welfare function

$$W(w_1, w_2, \dots, w_n) = \frac{1}{1 - \gamma} \left(w_1^{1-\gamma} + w_2^{1-\gamma} + \dots + w_n^{1-\gamma} \right).$$

If we think we should give no priority to the worse-off, then we can set $\gamma = 0$: the entire equation is then just a sum of welfare levels, which is the utilitarian social welfare function. By contrast, if we were maximally prioritarian, taking the limit of γ as it gets infinitely large, then we recover Rawls' maximin function. Similarly, if we took the limit of γ as it approached 1, we would recover the logarithmic social welfare function. This function is widespread in economic literature due to this versatility; among other things, it allows researchers to determine from observed preferences what a typical person's parameter γ is. By adopting this most general form, we can pick exactly how we want to prioritize individuals in society. This enables a principled way of trading off efficiency and equity.

Conclusions About Social Welfare Functions

Using AI to estimate social welfare. Artificial intelligence might be used to estimate the inputs and calculate the outputs of social welfare functions to help us decide between different actions and implement our ideal decision processes. By processing large amounts of data, AI systems might be able to predict individual welfare under various scenarios, paving the way for more informed decision-making. It has been demonstrated, for instance, that context-aware machine learning models like large language models can be used for predicting disease incidence faster than conventional epidemiological methods, which can be used to evaluate different public health proposals from a social welfare perspective. The application of AI can lead to more refined estimates of social welfare.

Using social welfare functions to train AI. A second connection between social welfare functions and AI lies in creating AI systems. We can shape AI systems to generally promote social welfare, ensuring they are aligned with human interests even when optimizing for distinct goals, such as corporate profit. Particularly in reinforcement learning, where decision-making is based on rewards or penalties, these can be tethered to societal wellbeing. Even for other agents, AI behavior can be guided towards promoting social welfare, aligning AI performance with societal wellbeing and making AI a compelling tool for social good.

Using AI to maximize social welfare. Beyond creating AI systems in line with social welfare, we can steer advanced AI to actively optimize social wellbeing by using social welfare functions as their objective functions. This harnesses AI's power to promote societal welfare effectively. Given AI's data processing and predictive capabilities, it can evaluate various strategies to identify what would best increase social welfare. By aligning AI objectives with social welfare functions, we can develop beneficial AI systems that not only recognize and understand social wellbeing but also actively work towards enhancing it. This proactive use of AI bolsters our ability to build societies where individual wellbeing is optimized and evenly distributed, reflecting our social preferences.

Challenges. Social welfare functions require choosing a theory of wellbeing. AI systems that maximize social welfare may also inherit various problems of current models such as vulnerability to adversarial attacks and proxy gaming and difficulties in interpreting the internal logic that led them to make certain decisions (both discussed in the Single Agent Safety chapter),

Summary. In this section, we discussed social welfare functions: mathematical functions that tell us how to aggregate information about individual welfare into one society-wide measure. We considered why social welfare functions might be relevant in the context of decision making by beneficial AI systems and explored a few properties that all social welfare functions possess. We analyzed how governments today use cost benefit analysis as an approximation to social welfare functions and why that falls short of what we desire. The main point of this section was to document and understand some of the most common social welfare functions – the utilitarian

function, the Rawlsian maximin function, and continuous prioritarian functions – and how they differ in their levels of concern for the well-being of different individuals, and can be represented within the framework of a general isoelastic social welfare function.

We think that the use of AI can help estimate social welfare inputs and calculate outputs, and AI systems can be shaped to generally promote social welfare and actively optimize it using social welfare functions as objective functions. Ultimately, the proactive use of AI bolsters our ability to build societies where individual well-being is both optimized and evenly distributed, reflecting our social preferences.

6.9 MORAL UNCERTAINTY

6.9.1 Making Decisions Under Moral Uncertainty

This section considers how we can make decisions when we are unsure which moral view is correct, and what this might imply for how we should design AI systems. Although ignoring our uncertainty may be a comfortable approach in daily life, there are situations where it is crucial to identify the best decision. We will start by considering our uncertainties about morality and the idea of reasonable pluralism, which acknowledges the potential co-existence of multiple reasonable moral theories such as ethical theories, common-sense morality, and religious teachings. We will explore uncertainties about moral truths, why they matter in moral decision-making for both humans and AI, and how to deal with them. We will look at a few proposed solutions, including *My Favorite Theory*, *Maximize Expected Choice-Worthiness (MEC)*, and *Moral Parliament* [351]. These approaches will be compared and evaluated in terms of their ability to help us make moral decisions under uncertainty. We will then briefly explore how we might use one of these solutions, the idea of a Moral Parliament, to enable AI systems to capture moral uncertainty in their decision-making.

Dealing with Moral Uncertainty

Moral theories Moral theories are systematic attempts to provide a general account of moral principles that apply universally. Good moral theories should provide a coherent, consistent framework for determining whether an action is right or wrong. A basic background understanding of some of the most commonly held moral theories provides a useful foundation for thinking about the kinds of goals or ideals that we wish AI systems to promote. Without this background, there is a risk that developers and users of AI systems may jump to conclusions about these topics with a false sense of certainty and without considering many potential considerations that could change their decisions. It would be highly inefficient for those developing AI systems or trying to make them safer to attempt to re-invent moral systems, without learning from the large existing body of philosophical work on these topics. We do not have space here to explore moral theories in depth, but we provide suggestions in the Recommended Reading section for those looking for a more detailed introduction to these topics.

There are many different types of moral theories, each of which emphasizes different moral values and considerations. Consequentialist theories like *utilitarianism* hold that the morality of an action is determined by its consequences or outcomes. Utilitarianism places an emphasis on maximizing everyone's wellbeing. Utilitarianism claims that consequences (and only consequences) determine whether an action is right or wrong, that wellbeing is the only intrinsic good, that everyone's wellbeing should be weighed impartially, and that we should maximize wellbeing.

By contrast, under deontological theories, some actions (like lying or killing) are simply wrong, and they cannot be justified by the good consequences that they might bring about. Deontology is the name for a family of ethical theories that deny that the rightness of actions is solely determined by their consequences. Deontological theories are systems of rules or obligations that constrain moral behavior [352]. The term *deontology* encompasses religious ethical theories, non-religious ethical theories, and principles and rules that are not part of theories at all. These theories give obligations and constraints priority over consequences.

Other moral theories may emphasize other values. For example, *social contract theory* (or contractarianism) focuses on contracts—or, more generally, hypothetical agreements between members of a society—as the foundation of ethics. A rule such as “do not kill” is morally right, according to a social contract theorist, because individuals would agree that the adoption of this rule is in their mutual best interest, and would therefore insert it into a social contract underpinning that society.

Moral uncertainty requires us to consider multiple moral theories. Individuals may have varying degrees of belief in different moral theories, known as *credence*.

Credence is the probability assigned by an individual of the chance a theory is true. Someone may have a high degree of credence (70%) in utilitarianism, meaning they believe that maximizing utility is likely to be the most important moral principle. However, they may also have some credence in deontological rules, believing they are plausible but less likely to be true than utilitarianism (30%). Since ethical theories are not flawless, often arriving at intuitively questionable conclusions, it is valuable to consider multiple perspectives when making moral decisions.

Living a good life can require a combination of insights from multiple moral theories. We often find ourselves balancing different kinds of deeds: creating pleasure for people, respecting autonomy, and adhering to societal moral norms. Abiding by a *reasonable pluralism* means accepting that moral guidance from different sources may conflict while still offering value.

In high-stakes moral decisions, such as in healthcare or AI design, reasonable pluralism alone may fall short. When a healthcare administrator faces the tough choice of allocating limited resources between hospitals, we might want them to seriously consider the ethics of what they are doing rather than just go with conflicting wisdom that seems reasonable at first glance. Therefore, we must

think hard about how to make decisions under moral uncertainty—seeking truth for crucial decisions is vital.

If AI is unable to account for moral uncertainty, harmful outcomes are likely. As AI systems become increasingly advanced, they are likely to become better at optimising for the goals that we set them, finding more creative and powerful solutions to achieve these. However, if they pursue very narrowly specified goals, there is a risk that these solutions come at the expense of other important values that we failed to adequately include as part of their goals.

Moral disagreement is a reason for making AI systems cautious. Given that philosophers have not yet converged on a single consistent moral theory that can take account of all relevant arguments and intuitions, it seems important for us to design AI systems without encoding a false sense of certainty about moral issues that could lead to unintended consequences. To counter such problems, we need AI systems to recognize that a broad range of moral perspectives might be valid—in other words, we need AI systems to acknowledge moral uncertainty. This presents another challenge: how should we rationally balance the recommendations of different ethical theories? This is the question of moral uncertainty.

How Should We Approach Moral Uncertainty?

There are several potential solutions to moral uncertainty. Faced with ethical uncertainty, we can turn to systematic approaches, using our estimates of how theories judge different actions and how likely these theories are. This section explores three potential solutions to moral uncertainty. The first is adopting a favored moral theory that aligns with personal beliefs (*My Favorite Theory*). The second is aiming for the highest average moral value by calculating the expected choice-worthiness of each option (*Maximize Expected Choice-Worthiness*). The third is treating the decision as a negotiation in a parliament, considering multiple moral views to find a mutually acceptable solution (*Moral Parliament*).

Consider whether we should lie to save a life. Imagine that a notorious murderer asks Alex where his friend, Jordan, is. Alex knows that revealing Jordan's location will likely lead to his friend's death, while lying would save Jordan's life. However, lying is morally questionable. Alex must decide which action to take. He is unsure, and considers the recommendations of the three moral theories he has some credence in: utilitarianism, deontology, and contractarianism. Alex thinks utilitarianism, which values lying to save a life highly, is the most likely to be true: he has 60% credence in it. Deontology, which Alex has 30% credence in, strongly disapproves of lying, even to save a life, and contractarianism, which Alex has 10% credence in, moderately approves of lying in this situation. This information is represented in Table 6.4.

Under My Favorite Theory (MFT), Alex would pick whatever utilitarianism recommends. Alex thinks that utilitarianism is the most likely to be true. The MFT approach is to follow the prescription of the theory we believe is the closest

TABLE 6.4. Example: Alex’s credence in various theories and their evaluation of lying to save a life.

	Utilitarianism	Deontology	Contractarianism
What is Alex’s estimate of the chance this theory is true	60%	30%	10%
Does this theory like lying to save a life?	Yes	No	Yes

to a moral truth. Intuitively, many people do this already when thinking about morality. The advantage of MFT is its simplicity: it is relatively simple and straightforward to implement. It does not require complex calculations or a detailed understanding of different moral perspectives. This approach can be useful when the level of moral uncertainty is low, and it is clear which theory or option is the best choice.

However, following MFT can lead to harmful single-mindedness or over-confidence. It can be difficult to put aside personal biases or to recognize when one’s own moral beliefs are fallible (individuals tend to defend and rationalize their chosen theories). The key issue with MFT is that it can discard relevant information, such as when the credences in two theories are close, but their judgments of an action vastly differ [353]. Imagine having 51% credence in contractarianism, which mildly supports lying to save a life, and 49% credence in Deontology, which views it as profoundly immoral. MFT suggests following the marginally favored theory, even though the potential harm, according to the second theory, is much larger. This seems counterintuitive, indicating that MFT might not always provide the most sensible approach for navigating moral uncertainty.

Maximize Expected Choice-Worthiness (MEC) gives us a procedure to follow. MEC tells us that to determine how to act, we need to do the following two things:

1. **Determine choice-worthiness.** Choice-worthiness is a measure of the overall desirability or value of an option—in this context, it is how morally good a choice is. This is an expression of the size of the moral value of an action. We can represent the choice-worthiness of an action as a number. For instance, we might think that under utilitarianism, the choice-worthiness of murder could be -1000 , that of littering could be -2 , that of helping an old lady cross the street could be $+10$, and that of averting existential risk could be $+10000$.
2. **Multiply choice-worthiness by credence.** We can consider the average choice-worthiness of an action, weighted by how likely we think each theory is to be true. This gives us a sense of our best guess of the average moral value of an action, much like how we considered expected utility when discussing utilitarianism. Table 6.5 has each theory’s choice-worthiness value for lying to save a life. As before, utilitarianism highly values lying to save a life ($+500$), deontology strongly disapproves of it (-1000), and contractualism moderately approves of it ($+100$). In

the row beneath these values are the credence probability-weighted judgments for Alex. The total calculation is

$$\frac{60}{100} \cdot 500 + \frac{30}{100}(-1000) + \frac{10}{100} \cdot 10 = 300 - 300 + 10 = 10$$

Under MEC, Alex would choose to lie, because given Alex’s credence in each moral theory and his determination of how each moral theory judges lying to save a life, lying has a higher expected choice-worthiness. Alex would lie because he judges that, on average, lying is the best possible action.

TABLE 6.5. Example: Alex’s credence in various theories, their evaluation of lying to save a life, and their probability-weighted contribution to the final judgment.

	Utilitarianism	Deontology	Contractarianism
What is Alex’s estimate of the chance this theory is true	60%	30%	10%
How much does this theory like lying to save a life	+500	−1000	+100
What is the probability-weighted judgment?	+300	−300	+10

MEC gives us a way of balancing how likely we think each theory is with how much each theory cares about our actions. We can see that utilitarianism and deontology’s relative contributions to the total moral value cancel out, and we are left with an overall “+10” in favor of lying to save a life. This calculation tells us that—when accounting for how likely Alex thinks each theory is to be true and how strong the theories’ preferences over his actions are—Alex’s best guess is that this action is morally good. (Although, since these numbers are rough and the final margin is quite thin, we would be wary of being overconfident in this conclusion: the numbers do not necessarily represent anything true or precise.) MEC has distinct advantages. For instance, unlike MFT, we ensure that we avoid actions that we think are probably fine but might be terrible, since large negative choice-worthiness from some theories will outweigh small positives from others. This is a sensible route to take in the face of moral uncertainty.

However, MEC faces challenges when comparing theories. While some cases are neatly solved by MEC, its philosophical foundations are questionable. Our assignment of choice-worthiness can be arbitrary: virtue ethics, for instance, advises acting virtuously without clear guidelines. MEC is unable to deal with ‘ordinal’ theories that only rank actions by their moral value rather than explicitly judging how morally right or wrong they are, making it difficult to determine choice-worthiness. Absolutist theories, like extreme Kantian ethics, deem certain actions absolutely wrong. We had initially assigned −1000 for the value of lying to save a life for a deontologist; it is unclear what we could put that would capture such an absolutist view. If we considered this to be infinitely bad, which seems like an accurate representation of the view, then it would overwhelm any other non-absolutist theory.

Even if we think it is simply very large, these firm stances can be more forceful than other ethical viewpoints, despite ascribing a low probability to the theory's correctness. This is because even a small percentage of a large value is still meaningfully large; consider that 0.01% of 1,000,000 is still 100 – a figure that may outweigh other theories we deem more probable.

Following a moral parliament approach, Alex could consider the proposal to lie more thoroughly and make a considered decision. In the moral parliament, imagined delegates representing different moral theories negotiate to find the best solution. In Alex's moral parliament, there would be 60 utilitarian delegates, 30 deontological delegates, and 10 contractarian delegates—numbers proportional to his credence in each theory. These delegates would negotiate and then come to a final decision by voting. Drawing inspiration from political systems, the moral parliament allows an agent to find flexible recommendations that enable compromises among plausible theories.

Depending on the voting rule, moral parliament can lead to different outcomes. In a conventional setting with majority rule, the utilitarians in Alex's moral parliament would always be able to push through their decisions. To avoid such outcomes, philosophers recommend using *proportional chances voting*. Here, an action is taken with a probability equal to its vote-share: in this case, if no one changed their mind, then the outcome of the parliament would recommend with 70% probability that Alex lie and with 30% probability that he tells the truth, since only the 30 deontological delegates would vote against this proposal. This encourages negotiation even if there is already a majority, which naturally leads to more cooperative outcomes. This is a more intuitive approach than, for instance, assigning choice-worthiness values and multiplying things out, even if it does take more effort.

However, it is difficult to see what a moral parliament might recommend. We can envision a variety of different possible outcomes in Alex's case. We might have the simple outcome described above, where no one changes their mind. Or, we might think that the deontologists can convince the contractualists that lying is bad in this case because lying would not be tolerated behind the veil of ignorance, reducing the chance of lying to 60%. Or, the parliament might even propose something entirely new, such as a compromise in which Alex does not explicitly lie but simply omits the truth. All of these are reasonable—it is difficult to choose between them.

The outcome of the moral parliament is not determined externally. Instead, it is a matter of our imagination, subject to our biases. Often, individuals resist imagining things that contradict their preconceived notions. This means that moral parliament may not be very helpful for individuals thinking about what is right to do in practice. With enough resources, however, we might be able to simulate model parliaments to recommend such decisions for us, such as by hiring diplomats to represent moral positions and having them bargain—or by assigning moral views to multiple AI systems and having them come to a collective decision.

6.9.2 Implementing a Moral Parliament in AI Systems

AIs might use simulated moral parliaments to account for moral uncertainty. If we want AIs to be guided by moral theories while accounting for moral uncertainty, we might use AI systems to simulate a moral parliament. We could imagine using advanced AI systems to emulate these representatives by training AIs to act in accordance with a specific moral theory. This would enable us to run moral parliaments artificially, permitting real-time decision-making. Just like in real parliaments, we would have delegates—AIs instructed to represent certain moral theories—get together, discuss what to do, negotiate favorable outcomes, and then vote on what to recommend. We could use the output of this moral parliament to instruct a separate AI in the real world to take certain actions over others.

This is speculative and might still face problems; for instance, AIs might have insufficient understanding of our moral theories. However, these problems could become more tractable with advanced AI systems. Assuming we have this ability, using a moral parliament might be an attractive solution to getting AIs to act in accordance with human values in real time.

Moral parliaments could be useful for representing stakeholders, not just theories. While we have explored the traditional moral parliament method of representing moral theories, we can generalize beyond this. Instead of representing theories, it might be more appropriate to represent stakeholders; for instance, in a decision about public transport, we could emulate representatives for local residents, commuters, and environmental groups, all of whom have an interest in the outcome. Using a generalized moral parliament for decision-making in AI is an approach that ensures all relevant perspectives are taken into account. In contrast to traditional methods that focus on representing different moral theories, *stakeholder representation* prioritizes the views of those directly affected by the AI's decisions. This could enhance the AI's understanding of the intricate human social dynamics involved in any given situation.

6.9.3 Advantages of a Moral Parliament

Using moral parliaments presents a wide array of benefits relative to just giving AIs certain sets of values directly. In this subsection, we will explore how they are customizable, transparent, robust to bugs and errors, adaptable to changing human values, and pro-negotiation.

Customizable moral parliaments are diverse and scalable. The generalized moral parliament can accommodate a wide variety of stakeholders, ranging from individual users to large corporations, and from local communities to global societies. By emulating a large set of stakeholders, we can ensure that a diverse set of views are represented. This allows AIs to effectively respond to a wide range of scenarios and contexts, providing a robust framework for ensuring AI decisions reflect the values, interests, and expectations of all relevant stakeholders. Additionally, moral parliaments are scalable: if we are concerned about a lack of representation, we can simply

emulate more stakeholders. By grounding AI decision-making in human perspectives and experiences, we can create AI systems that are not only more ethical and fair but also more effective and beneficial for society as a whole.

Transparency is another key benefit of a moral parliament. As it stands, automated decision-making is opaque: we rarely understand why AIs make the decisions they do. However, since the moral parliament gives us a clear mechanism of representing and weighing different perspectives, it allows stakeholders to understand the basis of an AI's decision-making. We could, for instance, enforce that AIs keep records of simulated negotiations in human languages and then view the transcripts. Using moral parliaments provides insights into how different moral considerations have been weighed against each other, making the decision-making process of an AI more transparent, explainable, and accountable.

Moral parliaments may result in AI systems that are less fragile and less prone to over-confidence. If we are sure that utilitarianism is the correct moral view, we might be tempted to create AIs that maximize wellbeing—this seems clean and elegant. However, having a diverse moral parliament would make AIs less likely to misbehave. By having multiple parliament members, we would achieve *redundancy*. This is a common principle in engineering: to always include extra components that are not strictly necessary to functioning, in case of failure in other components (and is explored further in the Safety Engineering chapter). We would do this to avoid failure modes where we were overconfident that we knew the correct moral theory, such as lying and stealing for the greater good, or just to avoid poor implementation from AIs optimizing for one moral theory. For example, a powerful AI told that utilitarianism is correct might implement utilitarianism in a particular way that is likely to lead to bad outcomes. Imagine an AI that has to evaluate millions of possibilities for every decision it makes. Even with a small error rate, the cumulative effect could lead the AI to choose risky or unconventional actions. This is because, when evaluating so many options, actions with high variance in moral value estimation may occasionally appear to have significant positive value. The AI could be more inclined to select these high-risk actions based on the mistaken belief that they would yield substantial benefits. For instance, an AI following some form of utilitarianism might use many resources to create happy digital minds—at the expense of humanity—even if that is not what we humans think is morally good.

This is similar to the Winner's Curse in auction theory: those that win auctions of goods with uncertain value often find that they won because they overestimated the value of the good relative to everyone else; for instance, when bidding on a bag of coins at a fair, people who overestimate how many coins there are will be more likely to win. Similarly, the AI might opt for actions that, in hindsight, were not truly beneficial. A moral parliament can make this less likely, because actions that would be judged morally extreme by most humans also wouldn't be selected by a diverse moral parliament.

The process of considering a range of theories inherently embeds redundancy and cross-checking into the system, reducing the probability of catastrophic outcomes

arising from a single point of failure. It also helps ensure that AI systems are robust and resilient, capable of handling a broad array of ethical dilemmas.

Moral parliaments encourage compromise and negotiation. In real-life parliaments, representatives who hold different opinions on various issues often engage in bargaining, compromise, and cooperation to reach agreeable outcomes and find common ground. We want our AIs to achieve similar outcomes, such as ones that are moderate instead of extreme. Ideally, we want AIs to select outcomes that many moral theories and stakeholders all like, rather than being forced to trade off between them.

In particular, we might want to design our moral parliaments in specific ways to encourage this. One such feature is proportional chances voting, in which each option then gets a chance of winning that's proportional to the number of votes it gets—if a parliament is 60/40 split on a proposal, then the AI would do what's recommended 60% of the time rather than just going with the majority. This setup motivates the representatives to come together on options that are compromises rather than sticking to their own viewpoints rigidly. They want to do this to prevent any option they see as extremely bad from having any chance of winning. This ensures a robust high-level principle guiding AI behavior, reducing the risk of extreme outcomes, and fostering a more balanced, nuanced approach to ethical decision-making.

Using a moral parliament reduces the risk of overlooking or locking in certain values. The moral parliament represents an approach to ethical decision-making in AI that is distinctively cosmopolitan, in that it encompasses a broad range of moral theories or stakeholders. It ensures that many ethical viewpoints are considered. This wider view is helpful in dealing with moral problems and tough decisions AI systems may run into, by making sure that all important considerations are thought over in a balanced way. AIs using moral parliaments are less likely to ignore values that matter to different groups in society.

Further, the moral parliament allows the representation of human values to grow and change over time. We know that moral views change over time, so we should be humble about how much we know about morality: there might be important things we don't yet understand that could help us get closer to the truth about what is right and wrong. By regularly using moral parliaments, AI systems can keep up with current human values, rather than sticking to the old values that were defined when the AI was created. This keeps AI up-to-date and flexible, and prevents it from acting based on outdated or irrelevant values that are locked into the system.

Challenges. Deciding which ethical theories to include in the moral parliament could be a challenging task. There are numerous ethical frameworks, and selecting a representative set may be subjective and politically charged. The decision procedure used to assign appropriate weights to different ethical theories and aggregate their recommendations in ways that reflect their importance could also be contentious. Different stakeholders may have varying opinions on how to prioritize these theories. Moreover, ethical theories can be subject to interpretation and may have nuanced

implications. Advanced AI systems would need to be able to accurately understand and apply these theories in order to use a moral parliament.

Conclusions about Moral Uncertainty

Taking moral uncertainty into account is difficult but important. As AI systems become increasingly embedded across various parts of society and take more consequential decisions, it will become more important to ensure that they can handle moral uncertainty. The same AI systems may be used by a wide variety of people with different moral theories, who would demand that AI acts as far as possible in ways that do not violate their moral views. Incorporating moral uncertainty into AI decision-making could reduce the probability of taking actions that are seriously wrong under some moral theories.

In this section, we explored ways of moving beyond intuitive judgments or a reasonable pluralism of theories, examining three solutions to moral uncertainty: my favorite theory, maximize expected choice-worthiness, and moral parliament. Each approach has its strengths and limitations, with the choice depending on the situation, level of uncertainty, and personal preferences.

We also discussed how we might use AIs to operationalize moral parliaments, whether in the original form of representing moral theories or generalized to representing stakeholders for any given issue. We highlighted the advantages of using a moral parliament, such as reducing the risk of overlooking or locking in certain values, allowing for the representation of changing human values over time, and increasing transparency and accountability in AI decision-making. We also noted that moral parliaments encourage compromise and negotiation, leading to more balanced and nuanced ethical decisions.

It is important to recognize that there might not be a one-size-fits-all solution to ethical dilemmas. As AI technology progresses, we should work further to develop robust methods for addressing moral uncertainty, that can include diverse moral perspectives and quantify uncertainty.

6.10 CONCLUSION

Overview. In this chapter, we have explored various ways in which we can embed ethics into AI systems, ensuring that they are safe and beneficial. It is far from guaranteed that the development of AIs will lead to socially beneficial outcomes. By default, AIs are likely to be developed according to businesses' economic incentives and are likely to follow parts of the law. This is insufficient. We almost certainly need stronger protections in place to ensure that AIs behave ethically. Consequently, we discussed how we can ensure AIs prioritize aspects of our wellbeing by making us happy, helping us flourish, and satisfying our preferences. Supposing AIs can figure out how to promote individual wellbeing, we explored social welfare functions as a way to guide their actions in order to help improve wellbeing across society.

We should strive to make our views less contradictory. We have considered several possible perspectives on how to ensure AIs act ethically. As we have seen, there are deep tensions between many plausible views, such as “AIs should do what you choose” versus “AIs should do what you want” versus “AIs should do what makes you happy” and so on. It is quite difficult to resolve these tensions and choose how to best represent human values; however, before we deploy powerful AI systems, we must do so anyway.

As a baseline, we want AIs to follow the law. At the very least, we should require that AIs follow the law. This is imperfect: as we have seen, the law is insufficiently comprehensive to ensure that AI systems are safe and beneficial. Laws have loopholes, are occasionally unethical and unrepresentative of the population, and are often silent on doing good in ways AIs should be required to do. However, if we can get AIs to follow the law, then we are at least guaranteed that they refrain from the illegal acts—such as murder and theft—that human societies have identified and outlawed. In addition, we might want to support regulation that ensures that AI decision-making must be fair—once we understand what fairness requires.

AIs could increase human wellbeing in accordance with a social welfare function. Which conception of human wellbeing best matches reality and how we should distribute it are difficult questions that we have certainly not resolved within this chapter. However, if we had to guess, we might want AIs to optimize continuous prioritarian social welfare functions where individual wellbeing should be based on happiness or objective goods, ensuring that wellbeing is fairly distributed throughout a society in which everyone is happy to live. We might use AIs to estimate general-purpose wellbeing functions and directly increase what we observe makes people better off. While this is speculative, the rapid development of AIs forces us to speculate.

Further work is needed on how to embed ethics into AI systems. As we move forward, it is crucial that we continue to engage in rigorous research, open dialogue, and interdisciplinary collaboration to address the ethical concerns associated with AI. By doing so, we can strive towards creating AI systems that not only avoid the worst harms to society but actively work towards enhancing social wellbeing. In the next part of this book, we will move to the problem of how to ensure positive outcomes in a world with multiple AI agents and many, often competing, human stakeholders.

6.11 LITERATURE

6.11.1 Recommended Reading

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